1) a)

Geometric with common ratio $r = \frac{1}{3}$

1) b)

Not geometric. 5 terms so coefficients are $1, 4, 6, 4, 1$:

$1 + 0.8 + 0.24 + 0.032 + 0.0016$

$= 1 \times 1 + 4 \times 0.2 + 6 \times 0.04 + 4 \times 0.008 + 1 \times 0.0016$

$= 1 \times 1 + 4 \times 0.2 + 6 \times 0.2^2 + 4 \times 0.2^3 + 1 \times 0.2^4$

$= (1 + 0.2)^4$

So binomial series.

1) c)

Geometric with common ratio $r = -\frac{1}{4}$

1) d)

Not geometric. 4 terms so coefficients are $1, 3, 3, 1$:

$8 - 1.2 + 0.06 - 0.001 = 1 \times 8 + 3 \times -0.4 + 3 \times 0.02 + 1 \times -0.001$

$= 1 \times 2^3 + 3 \times 2^2 \times -0.1 + 3 \times 2 \times (-0.1)^2 + 1 \times (-0.1)^3$

$= (2 - 0.1)^3$

So binomial series.

2) a)

$a = 4$, $r = \frac{1}{3}\mathrm{e}^{\mathrm{i}\theta}$

$S_\infty = \frac{a}{1-r} = \frac{4}{1-\frac{1}{3}\mathrm{e}^{\mathrm{i}\theta}} = \frac{12}{3-\mathrm{e}^{\mathrm{i}\theta}}$

2) b)

$a = 1$, $r = \frac{1}{4}\mathrm{e}^{\mathrm{i}\theta}$

$S_\infty = \frac{a}{1-r} = \frac{1}{1-\frac{1}{4}\mathrm{e}^{\mathrm{i}\theta}} = \frac{4}{4-\mathrm{e}^{\mathrm{i}\theta}}$

2) c)

$a = 2, r = \frac{1}{5}e^{i2\theta}$

$S_\infty = \frac{a}{1-r} = \frac{2}{1-\frac{1}{5}e^{i2\theta}} = \frac{10}{5-e^{i2\theta}}$

2) d)

$a = 5, r = \frac{1}{10}e^{i3\theta}$

$S_\infty = \frac{a}{1-r} = \frac{5}{1-\frac{1}{10}e^{i3\theta}} = \frac{50}{10-e^{i3\theta}}$

2) e)

$a = 6, r = \frac{1}{12}e^{-i\theta}$

$S_\infty = \frac{a}{1-r} = \frac{6}{1-\frac{1}{12}e^{-i\theta}} = \frac{72}{12-e^{-i\theta}}$

2) f)

$a = 2, r = \frac{1}{6}e^{-i4\theta}$

$S_\infty = \frac{a}{1-r} = \frac{2}{1-\frac{1}{6}e^{-i4\theta}} = \frac{12}{6-e^{-i4\theta}}$

3) a)

$a = e^{i\theta}, r = e^{i\theta}, n = 12$

$S_n = \frac{a(1-r^n)}{1-r} = \frac{e^{i\theta}(1-e^{i12\theta})}{1-e^{i\theta}} = \frac{e^{i\theta}-e^{i13\theta}}{1-e^{i\theta}}$

3) b)

$a = 2, r = \frac{1}{2}e^{i\theta}, n = 10$

$S_n = \frac{a(1-r^n)}{1-r} = \frac{2(1-\frac{1}{2^{10}}e^{i10\theta})}{1-\frac{1}{2}e^{i\theta}} = \frac{2-\frac{2}{1024}e^{i10\theta}}{1-\frac{1}{2}e^{i\theta}} = \frac{1024-e^{i10\theta}}{512-256e^{i\theta}}$

3) c)

$a = 1, r = -\frac{1}{3}e^{i\theta}, n = 9$

$S_n = \frac{a(1-r^n)}{1-r} = \frac{1(1+\frac{1}{3^9}e^{i9\theta})}{1+\frac{1}{3}e^{i\theta}} = \frac{1+\frac{1}{19683}e^{i9\theta}}{1+\frac{1}{3}e^{i\theta}} = \frac{19683+e^{i9\theta}}{19683+6561e^{i\theta}}$

3) d)

$a = 16, r = -\frac{1}{4}\mathrm{e}^{\mathrm{i}2\theta}, n = 8$

$S_n = \frac{a(1-r^n)}{1-r} = \frac{16(1-\frac{1}{4^8}\mathrm{e}^{\mathrm{i}16\theta})}{1+\frac{1}{4}\mathrm{e}^{\mathrm{i}2\theta}} = \frac{16-\frac{1}{4096}\mathrm{e}^{\mathrm{i}16\theta}}{1+\frac{1}{4}\mathrm{e}^{\mathrm{i}2\theta}} = \frac{65536-\mathrm{e}^{\mathrm{i}16\theta}}{4096+1024\mathrm{e}^{\mathrm{i}2\theta}}$

4) a)

$z^* = a - \mathrm{i}b$

4) b)

$z^* = r(\cos\theta - \mathrm{i}\sin\theta) = r(\cos(-\theta) + \mathrm{i}\sin(-\theta))$

4) c)

$z^* = r\mathrm{e}^{-\mathrm{i}\theta}$

5) a)

$z^* = 5 + \mathrm{e}^{-\mathrm{i}\theta}$

5) b)

$z^* = 3 - \mathrm{e}^{\mathrm{i}\theta}$

5) c)

$z^* = 7 + \mathrm{e}^{-\mathrm{i}2\theta}$

5) d)

$z^* = 7 + 9\mathrm{e}^{-\mathrm{i}\theta}$

5) e)

$z^* = 2 - 3\mathrm{e}^{-\mathrm{i}\theta}$

5) f)

$z^* = 1 + \mathrm{e}^{\mathrm{i}\theta}$

5) g)

$z^* = 2 - \mathrm{e}^{-\mathrm{i}\theta}$

5) h)

$z^* = 5 - 5\mathrm{e}^{\mathrm{i}\theta}$

5) i)

$z^* = 6 - 8\mathrm{e}^{-\mathrm{i}7\theta}$

5) j)

$z^* = 2 + 15\mathrm{e}^{-\mathrm{i}\frac{1}{2}\theta}$

6) a)

$\left(1 + 11\mathrm{e}^{-\mathrm{i}\theta}\right)\left(1 + 11\mathrm{e}^{\mathrm{i}\theta}\right) = 1 + 11\mathrm{e}^{\mathrm{i}\theta} + 11\mathrm{e}^{-\mathrm{i}\theta} + 121\mathrm{e}^{-\mathrm{i}\theta}\mathrm{e}^{\mathrm{i}\theta}$

$= 122 + 11\mathrm{e}^{\mathrm{i}\theta} + 11\mathrm{e}^{-\mathrm{i}\theta}$

6) b)

$\left(4 + \mathrm{e}^{\mathrm{i}3\theta}\right)\left(4 + \mathrm{e}^{-\mathrm{i}3\theta}\right) = 16 + 4\mathrm{e}^{\mathrm{i}3\theta} + 4\mathrm{e}^{-\mathrm{i}3\theta} + \mathrm{e}^{\mathrm{i}3\theta}\mathrm{e}^{-\mathrm{i}3\theta}$

$= 17 + 4\mathrm{e}^{\mathrm{i}3\theta} + 4\mathrm{e}^{-\mathrm{i}3\theta}$

6) c)

$\left(2 + \frac{1}{2}\mathrm{e}^{\mathrm{i}\theta}\right)\left(2 + \frac{1}{2}\mathrm{e}^{-\mathrm{i}\theta}\right) = 4 + \mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{-\mathrm{i}\theta} + \frac{1}{4}\mathrm{e}^{\mathrm{i}\theta}\mathrm{e}^{-\mathrm{i}\theta}$

$= \frac{17}{4} + \mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{-\mathrm{i}\theta}$

7) a)

$z = \cos\theta + \mathrm{i}\sin\theta$

7) b)

$\mathrm{e}^{\mathrm{i}\theta} = \cos\theta + \mathrm{i}\sin\theta$ and $\mathrm{e}^{-\mathrm{i}\theta} = \cos\theta - \mathrm{i}\sin\theta$

$\mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{-\mathrm{i}\theta} = \cos\theta + \mathrm{i}\sin\theta + \cos\theta - \mathrm{i}\sin\theta$

$= 2\cos\theta$

$\therefore \cos\theta = \frac{\mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{-\mathrm{i}\theta}}{2}$

8) a)

$10 + 6e^{i\theta} + 6e^{-i\theta} = 10 + 6\left(e^{i\theta} + e^{-i\theta}\right)$

$= 10 + 6(2\cos\theta)$

$= 10 + 12\cos\theta$

8) b)

$5 - 2e^{i4\theta} - 2e^{-i4\theta} = 5 - 2\left(e^{i4\theta} + e^{-i4\theta}\right)$

$= 5 - 2(2\cos 4\theta)$

$= 5 - 4\cos 4\theta$

8) c)

$\left(1 + e^{i3\theta}\right)\left(1 + e^{-i3\theta}\right) = 1 + e^{i3\theta} + e^{-i3\theta} + e^{i3\theta}e^{-i3\theta}$

$= 1 + \left(e^{i3\theta} + e^{-i3\theta}\right) + 1$

$= 2 + 2\cos 3\theta$

8) d)

$\left(2 + e^{i2\theta}\right)\left(2 + e^{-i2\theta}\right) = 4 + 2e^{i2\theta} + 2e^{-i2\theta} + e^{i2\theta}e^{-i2\theta}$

$= 4 + 2\left(e^{i2\theta} + e^{-i2\theta}\right) + 1$

$= 4 + 2(2\cos 2\theta) + 1$

$= 5 + 4\cos 2\theta$

8) e)

$\left(1 - \frac{1}{2}e^{i\frac{1}{3}\theta}\right)\left(1 - \frac{1}{2}e^{i\frac{1}{3}\theta}\right) = 1 - \frac{1}{2}e^{i\frac{1}{3}\theta} - \frac{1}{2}e^{-i\frac{1}{3}\theta} + \frac{1}{4}e^{i\frac{1}{3}\theta}e^{-i\frac{1}{3}\theta}$

$= 1 - \frac{1}{2}\left(e^{i\frac{1}{3}\theta} + e^{-i\frac{1}{3}\theta}\right) + \frac{1}{4}$

$= 1 - \frac{1}{2}\left(2\cos\frac{1}{3}\theta\right) + \frac{1}{4}$

$= \frac{5}{4} - \cos\frac{1}{3}\theta$

9) a)

$\frac{1}{2-e^{i\theta}} \times \frac{2-e^{-i\theta}}{2-e^{-i\theta}} = \frac{2-e^{-i\theta}}{4-2e^{i\theta}-2e^{-i\theta}+1}$

$= \frac{2-e^{-i\theta}}{4-2(e^{i\theta}+e^{-i\theta})+1}$

$= \frac{2-e^{-i\theta}}{5-4\cos\theta}$

9) b)

$$\frac{3}{4+e^{i2\theta}} \times \frac{4+e^{-i2\theta}}{4+e^{-i2\theta}} = \frac{12+3e^{-i2\theta}}{16+4e^{i2\theta}+4e^{-i2\theta}+1}$$
$$= \frac{12+3e^{-i2\theta}}{16+4(e^{i2\theta}+e^{-i2\theta})+1}$$
$$= \frac{12+3e^{-i2\theta}}{17+8\cos 2\theta}$$

9) c)

$$\frac{5}{6-2e^{-i\theta}} \times \frac{6-2e^{i\theta}}{6-2e^{i\theta}} = \frac{30-10e^{i\theta}}{36-12e^{i\theta}-12e^{-i\theta}+4}$$
$$= \frac{30-10e^{i\theta}}{40-12(e^{i\theta}+e^{-i\theta})}$$
$$= \frac{30-10e^{i\theta}}{40-24\cos\theta}$$
$$= \frac{15-5e^{i\theta}}{20-12\cos\theta}$$

9) d)

$$\frac{e^{i\theta}}{1-\frac{1}{3}e^{i3\theta}} \times \frac{1-\frac{1}{3}e^{-i3\theta}}{1-\frac{1}{3}e^{-i3\theta}} = \frac{e^{i\theta}-\frac{1}{3}e^{-i2\theta}}{1-\frac{1}{3}e^{i3\theta}-\frac{1}{3}e^{-i3\theta}+\frac{1}{9}}$$
$$= \frac{e^{i\theta}-\frac{1}{3}e^{-i2\theta}}{\frac{10}{9}-\frac{1}{3}(e^{i3\theta}+e^{-i3\theta})}$$
$$= \frac{e^{i\theta}-\frac{1}{3}e^{-i2\theta}}{\frac{10}{9}-\frac{2}{3}\cos 3\theta}$$
$$= \frac{9e^{i\theta}-3e^{-i2\theta}}{10-6\cos 3\theta}$$

9) e)

$$\frac{e^{i\theta}}{1+\frac{1}{8}e^{i4\theta}} \times \frac{1+\frac{1}{8}e^{-i4\theta}}{1+\frac{1}{8}e^{-i4\theta}} = \frac{e^{i\theta}+\frac{1}{8}e^{-i3\theta}}{1+\frac{1}{8}e^{i4\theta}+\frac{1}{8}e^{-i4\theta}+\frac{1}{64}}$$
$$= \frac{e^{i\theta}+\frac{1}{8}e^{-i3\theta}}{\frac{65}{64}+\frac{1}{8}(e^{i4\theta}+e^{-i4\theta})}$$
$$= \frac{e^{i\theta}+\frac{1}{8}e^{-i3\theta}}{\frac{65}{64}+\frac{2}{8}\cos 4\theta}$$
$$= \frac{64e^{i\theta}+8e^{-i3\theta}}{65+16\cos 4\theta}$$

10)

$$C + \mathrm{i}S = \cos\theta + \frac{1}{4}\cos 3\theta + \frac{1}{16}\cos 5\theta + \mathrm{i}\left(\sin\theta + \frac{1}{4}\sin 3\theta + \frac{1}{16}\sin 5\theta\right)$$
$$= (\cos\theta + \mathrm{i}\sin\theta) + \frac{1}{4}(\cos 3\theta + \mathrm{i}\sin 3\theta) + \frac{1}{16}(\cos 5\theta + \mathrm{i}\sin 5\theta)$$
$$= \mathrm{e}^{\mathrm{i}\theta} + \frac{1}{4}\mathrm{e}^{\mathrm{i}3\theta} + \frac{1}{16}\mathrm{e}^{\mathrm{i}5\theta}$$

11)

$$C + \mathrm{i}S = 1 - \frac{1}{2}\cos 2\theta + \frac{1}{4}\cos 4\theta + \mathrm{i}\left(\sin\theta - \frac{1}{2}\sin 2\theta + \frac{1}{4}\sin 4\theta\right)$$
$$= 1 - \frac{1}{2}(\cos 2\theta + \mathrm{i}\sin 2\theta) + \frac{1}{4}(\cos 4\theta + \mathrm{i}\sin 4\theta)$$
$$= \mathrm{e}^{0} - \frac{1}{2}\mathrm{e}^{\mathrm{i}2\theta} + \frac{1}{4}\mathrm{e}^{\mathrm{i}4\theta}$$

12) a)

$$\left(1 + \frac{1}{3}\mathrm{e}^{\mathrm{i}3\theta}\right)\left(1 + \frac{1}{3}\mathrm{e}^{-\mathrm{i}3\theta}\right) = 1 + \frac{1}{3}\mathrm{e}^{\mathrm{i}3\theta} + \frac{1}{3}\mathrm{e}^{-\mathrm{i}3\theta} + \frac{1}{9}\mathrm{e}^{\mathrm{i}3\theta}\mathrm{e}^{-\mathrm{i}3\theta}$$
$$= 1 + \frac{1}{3}\left(\mathrm{e}^{\mathrm{i}3\theta} + \mathrm{e}^{-\mathrm{i}3\theta}\right) + \frac{1}{9}$$
$$= 1 + \frac{1}{3}(2\cos 3\theta) + \frac{1}{9}$$
$$= \frac{10}{9} + \frac{2}{3}\cos 3\theta$$

12) b)

$$C + \mathrm{i}S = \cos\theta - \frac{1}{3}\cos 4\theta + \frac{1}{9}\cos 7\theta - \cdots + \mathrm{i}\left(\sin\theta - \frac{1}{3}\sin 4\theta + \frac{1}{9}\sin 7\theta - \cdots\right)$$
$$= (\cos\theta + \mathrm{i}\sin\theta) - \frac{1}{3}(\cos 4\theta + \mathrm{i}\sin 4\theta) + \frac{1}{9}(\cos 7\theta + \mathrm{i}\sin 7\theta) - \cdots$$
$$= \mathrm{e}^{\mathrm{i}\theta} - \frac{1}{3}\mathrm{e}^{\mathrm{i}4\theta} + \frac{1}{9}\mathrm{e}^{\mathrm{i}7\theta} - \cdots$$
$$a = \mathrm{e}^{\mathrm{i}\theta},\ r = -\frac{1}{3}\mathrm{e}^{\mathrm{i}3\theta}$$
$$S_\infty = \frac{a}{1-r} = \frac{\mathrm{e}^{\mathrm{i}\theta}}{1+\frac{1}{3}\mathrm{e}^{\mathrm{i}3\theta}}$$

12) c)

$$C + \mathrm{i}S = \frac{\mathrm{e}^{\mathrm{i}\theta}}{1+\frac{1}{3}\mathrm{e}^{\mathrm{i}3\theta}} \times \frac{1+\frac{1}{3}\mathrm{e}^{-\mathrm{i}3\theta}}{1+\frac{1}{3}\mathrm{e}^{-\mathrm{i}3\theta}} = \frac{\mathrm{e}^{\mathrm{i}\theta}+\frac{1}{3}\mathrm{e}^{-\mathrm{i}2\theta}}{\frac{10}{9}+\frac{2}{3}\cos 3\theta}$$

$$= \frac{9\mathrm{e}^{\mathrm{i}\theta}+3\mathrm{e}^{-\mathrm{i}2\theta}}{10+6\cos 3\theta}$$

$$= \frac{9(\cos\theta+\mathrm{i}\sin\theta)+3(\cos(-2\theta)+\mathrm{i}\sin(-2\theta))}{10+6\cos 3\theta}$$

$$= \frac{9\cos\theta+\mathrm{i}(9\sin\theta)+3\cos(2\theta)-\mathrm{i}(3\sin(2\theta))}{10+6\cos 3\theta}$$

$$= \frac{9\cos\theta+3\cos(2\theta)+\mathrm{i}(9\sin\theta-3\sin(2\theta))}{10+6\cos 3\theta}$$

Compare reals:

$$C = \frac{9\cos\theta+3\cos(2\theta)}{10+6\cos 3\theta}$$

12) d)

Compare imaginary:

$$S = \frac{9\sin\theta-3\sin(2\theta)}{10+6\cos 3\theta}$$

13) a)

$C + \mathrm{i}S = \cos 3\theta + \frac{1}{2}\cos 5\theta + \frac{1}{4}\cos 7\theta + \cdots + \mathrm{i}\left(\sin 3\theta + \frac{1}{2}\sin 5\theta + \frac{1}{4}\sin 7\theta + \cdots\right)$

$= (\cos 3\theta + \mathrm{i}\sin 3\theta) + \frac{1}{2}(\cos 5\theta + \mathrm{i}\sin 5\theta) + \frac{1}{4}(\cos 7\theta + \mathrm{i}\sin 7\theta) - \cdots$

$= \mathrm{e}^{\mathrm{i}3\theta} + \frac{1}{2}\mathrm{e}^{\mathrm{i}5\theta} + \frac{1}{4}\mathrm{e}^{\mathrm{i}7\theta} - \cdots$

$a = \mathrm{e}^{\mathrm{i}3\theta}, r = \frac{1}{2}\mathrm{e}^{\mathrm{i}2\theta}$

$S_\infty = \frac{a}{1-r} = \frac{\mathrm{e}^{\mathrm{i}3\theta}}{1-\frac{1}{2}\mathrm{e}^{\mathrm{i}2\theta}}$

$C + \mathrm{i}S = \frac{\mathrm{e}^{\mathrm{i}3\theta}}{1-\frac{1}{2}\mathrm{e}^{\mathrm{i}2\theta}} \times \frac{1-\frac{1}{2}\mathrm{e}^{-\mathrm{i}2\theta}}{1-\frac{1}{2}\mathrm{e}^{-\mathrm{i}2\theta}} = \frac{\mathrm{e}^{\mathrm{i}3\theta}-\frac{1}{2}\mathrm{e}^{\mathrm{i}\theta}}{1-\frac{1}{2}\mathrm{e}^{\mathrm{i}2\theta}-\frac{1}{2}\mathrm{e}^{-\mathrm{i}2\theta}+\frac{1}{4}}$

$= \frac{\mathrm{e}^{\mathrm{i}3\theta}-\frac{1}{2}\mathrm{e}^{\mathrm{i}\theta}}{\frac{5}{4}-\frac{1}{2}\left(\mathrm{e}^{\mathrm{i}2\theta}+\mathrm{e}^{-\mathrm{i}2\theta}\right)}$

$= \frac{\mathrm{e}^{\mathrm{i}3\theta}-\frac{1}{2}\mathrm{e}^{\mathrm{i}\theta}}{\frac{5}{4}-\cos 2\theta}$

$= \frac{4\mathrm{e}^{\mathrm{i}3\theta}-2\mathrm{e}^{\mathrm{i}\theta}}{5-4\cos 2\theta}$

$= \frac{4\cos 3\theta+4\mathrm{i}\sin 3\theta-2(\cos\theta+\mathrm{i}\sin\theta)}{5-4\cos 2\theta}$

$= \frac{4\cos 3\theta-2\cos\theta+\mathrm{i}(4\sin 3\theta-2\sin\theta)}{5-4\cos 2\theta}$

Compare reals:

$C = \frac{4\cos 3\theta-2\cos\theta}{5-4\cos 2\theta}$

13) b)

Compare imaginary:

$S = \frac{4\sin 3\theta-2\sin\theta}{5-4\cos 2\theta}$

14) a)

$C + \mathrm{i}S = 3 + \cos\theta + \frac{1}{3}\cos 2\theta + \cdots + \mathrm{i}\left(\sin\theta + \frac{1}{3}\sin 2\theta + \cdots\right)$

$= 3 + (\cos\theta + \mathrm{i}\sin\theta) + \frac{1}{3}(\cos 2\theta + \mathrm{i}\sin 2\theta) - \cdots$

$= 3 + \mathrm{e}^{\mathrm{i}\theta} + \frac{1}{3}\mathrm{e}^{\mathrm{i}2\theta} - \cdots$

$a = 3,\ r = \frac{1}{3}\mathrm{e}^{\mathrm{i}\theta}$

$S_\infty = \frac{a}{1-r} = \frac{3}{1-\frac{1}{3}\mathrm{e}^{\mathrm{i}\theta}}$

$C + \mathrm{i}S = \frac{3}{1-\frac{1}{3}\mathrm{e}^{\mathrm{i}\theta}} \times \frac{1-\frac{1}{3}\mathrm{e}^{-\mathrm{i}\theta}}{1-\frac{1}{3}\mathrm{e}^{-\mathrm{i}\theta}} = \frac{3-\mathrm{e}^{-\mathrm{i}\theta}}{1-\frac{1}{3}\mathrm{e}^{\mathrm{i}\theta}-\frac{1}{3}\mathrm{e}^{-\mathrm{i}\theta}+\frac{1}{9}}$

$= \frac{3-\mathrm{e}^{-\mathrm{i}\theta}}{\frac{10}{9}-\frac{1}{3}\left(\mathrm{e}^{\mathrm{i}\theta}+\mathrm{e}^{-\mathrm{i}\theta}\right)}$

$= \frac{3-\mathrm{e}^{-\mathrm{i}\theta}}{\frac{10}{9}-\frac{2}{3}\cos\theta}$

$= \frac{27-9\mathrm{e}^{-\mathrm{i}\theta}}{10-6\cos\theta}$

$= \frac{27-9(\cos(-\theta)+\mathrm{i}\sin(-\theta))}{10-6\cos\theta}$

$= \frac{27-9(\cos\theta-\mathrm{i}\sin\theta)}{10-6\cos\theta}$

$= \frac{27-9\cos\theta+\mathrm{i}(9\sin\theta)}{10-6\cos\theta}$

Compare reals:

$C = \frac{27-9\cos\theta}{10-6\cos\theta}$

14) b)

Compare imaginary:

$S = \frac{9\sin\theta}{10-6\cos\theta}$

15) a)

$\left(1-\frac{1}{2}\mathrm{e}^{\mathrm{i}2\theta}\right)\left(1-\frac{1}{2}\mathrm{e}^{-\mathrm{i}2\theta}\right) = 1-\frac{1}{2}\mathrm{e}^{\mathrm{i}2\theta}-\frac{1}{2}\mathrm{e}^{-\mathrm{i}2\theta}+\frac{1}{4} = \frac{5}{4}-\frac{1}{2}\left(\mathrm{e}^{\mathrm{i}2\theta}+\mathrm{e}^{-\mathrm{i}2\theta}\right)$

$= \frac{5}{4} - \cos 2\theta$ as req.

15) b)

$C + \mathrm{i}S = 2 + \cos 2\theta + \frac{1}{2}\cos 4\theta + \cdots + \frac{1}{512}\cos 18\theta + \mathrm{i}\left(\sin 2\theta + \frac{1}{2}\sin 6\theta + \cdots + \frac{1}{512}\sin 18\theta\right)$

$= 2 + (\cos 2\theta + \mathrm{i}\sin 2\theta) + \frac{1}{2}(\cos 4\theta + \mathrm{i}\sin 4\theta) + \cdots + \frac{1}{256}(\cos 18\theta + \mathrm{i}\sin 18\theta)$

$= 2 + \mathrm{e}^{\mathrm{i}2\theta} + \frac{1}{2}\mathrm{e}^{\mathrm{i}4\theta} + \cdots + \frac{1}{256}\mathrm{e}^{\mathrm{i}18\theta}$

$a = 2,\ r = \frac{1}{2}\mathrm{e}^{\mathrm{i}2\theta},\ n = 10$

$S_n = \frac{a(1-r^n)}{1-r} = \frac{2\left(1-\frac{1}{2^{10}}\mathrm{e}^{\mathrm{i}20\theta}\right)}{1-\frac{1}{2}\mathrm{e}^{\mathrm{i}2\theta}} = \frac{2-\frac{1}{512}\mathrm{e}^{\mathrm{i}20\theta}}{1-\frac{1}{2}\mathrm{e}^{\mathrm{i}2\theta}}$

$C + \mathrm{i}S = \frac{2-\frac{1}{512}\mathrm{e}^{\mathrm{i}20\theta}}{1-\frac{1}{2}\mathrm{e}^{\mathrm{i}2\theta}} \times \frac{1-\frac{1}{2}\mathrm{e}^{-\mathrm{i}2\theta}}{1-\frac{1}{2}\mathrm{e}^{-\mathrm{i}2\theta}}$

$= \frac{2-\mathrm{e}^{-\mathrm{i}2\theta}-\frac{1}{512}\mathrm{e}^{\mathrm{i}20\theta}+\frac{1}{1024}\mathrm{e}^{\mathrm{i}18\theta}}{\frac{5}{4}-\cos 2\theta}$

$= \frac{8-4\mathrm{e}^{-\mathrm{i}2\theta}-\frac{1}{128}\mathrm{e}^{\mathrm{i}20\theta}+\frac{1}{256}\mathrm{e}^{\mathrm{i}18\theta}}{5-4\cos 2\theta}$

$= \frac{2048-1024\mathrm{e}^{-\mathrm{i}2\theta}-2\mathrm{e}^{\mathrm{i}20\theta}+\mathrm{e}^{\mathrm{i}18\theta}}{256(5-4\cos 2\theta)}$

$= \frac{2048-1024(\cos(-2\theta)+\mathrm{i}\sin(-2\theta))-2(\cos 20\theta+\mathrm{i}\sin 20\theta)+1(\cos 18\theta+\mathrm{i}\sin 18\theta)}{5-4\cos 2\theta}$

$= \frac{2048-1024(\cos 2\theta-\mathrm{i}\sin 2\theta)-2(\cos 20\theta+\mathrm{i}\sin 20\theta)+1(\cos 18\theta+\mathrm{i}\sin 18\theta)}{5-4\cos 2\theta}$

$= \frac{2048-1024\cos 2\theta-2\cos 20\theta+\cos 18\theta+\mathrm{i}(1024\sin 2\theta-2\sin 20\theta+\sin 18\theta)}{5-4\cos 2\theta}$

Compare reals:

$C = \frac{2048-1024\cos 2\theta-2\cos 20\theta+\cos 18\theta}{256(5-4\cos 2\theta)}$

as req.

15) c)

Compare imaginary:

$S = \frac{1024\sin 2\theta-2\sin 20\theta+\sin 18\theta}{256(5-4\cos 2\theta)}$

as req.

16) a)

$\left(1 - \mathrm{e}^{\mathrm{i}2\theta}\right)\left(1 - \mathrm{e}^{-\mathrm{i}2\theta}\right) = 1 - \mathrm{e}^{\mathrm{i}2\theta} - \mathrm{e}^{-\mathrm{i}2\theta} + 1 = 2 - \left(\mathrm{e}^{\mathrm{i}2\theta} + \mathrm{e}^{-\mathrm{i}2\theta}\right) = 2 - 2\cos 2\theta$

16) b)

$C + \mathrm{i}S = \cos\theta + \cos 3\theta + \cos 5\theta + \cdots + \cos 45\theta + \mathrm{i}(\sin\theta + \sin 3\theta + \sin 5\theta + \cdots + \sin 45\theta)$

$= (\cos\theta + \mathrm{i}\sin\theta) + (\cos 3\theta + \mathrm{i}\sin 3\theta) + \cdots + (\cos 45\theta + \mathrm{i}\sin 45\theta)$

$= \mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{\mathrm{i}3\theta} + \mathrm{e}^{\mathrm{i}5\theta} + \cdots + \mathrm{e}^{\mathrm{i}45\theta}$

$a = \mathrm{e}^{\mathrm{i}\theta}, r = \mathrm{e}^{\mathrm{i}2\theta}, n = 23$

$S_n = \frac{a(1-r^n)}{1-r} = \frac{\mathrm{e}^{\mathrm{i}\theta}\left(1-\mathrm{e}^{\mathrm{i}46\theta}\right)}{1-\mathrm{e}^{\mathrm{i}2\theta}} = \frac{\mathrm{e}^{\mathrm{i}\theta}-\mathrm{e}^{\mathrm{i}47\theta}}{1-\mathrm{e}^{\mathrm{i}2\theta}}$

$C + \mathrm{i}S = \frac{\mathrm{e}^{\mathrm{i}\theta}-\mathrm{e}^{\mathrm{i}47\theta}}{1-\mathrm{e}^{\mathrm{i}2\theta}} \times \frac{1-\mathrm{e}^{-\mathrm{i}2\theta}}{1-\mathrm{e}^{-\mathrm{i}2\theta}} = \frac{\mathrm{e}^{\mathrm{i}\theta}-\mathrm{e}^{\mathrm{i}47\theta}-\left(\mathrm{e}^{-\mathrm{i}\theta}-\mathrm{e}^{\mathrm{i}45\theta}\right)}{1-\mathrm{e}^{\mathrm{i}2\theta}-\mathrm{e}^{-\mathrm{i}2\theta}+1}$

$= \frac{\mathrm{e}^{\mathrm{i}\theta}-\mathrm{e}^{\mathrm{i}47\theta}-\mathrm{e}^{-\mathrm{i}\theta}+\mathrm{e}^{\mathrm{i}45\theta}}{2-\left(\mathrm{e}^{\mathrm{i}2\theta}+\mathrm{e}^{-\mathrm{i}2\theta}\right)}$

$= \frac{\mathrm{e}^{\mathrm{i}\theta}-\mathrm{e}^{\mathrm{i}47\theta}-\mathrm{e}^{-\mathrm{i}\theta}+\mathrm{e}^{\mathrm{i}45\theta}}{2-2\cos 2\theta}$

$= \frac{(\cos\theta+\mathrm{i}\sin\theta)-(\cos 47\theta+\mathrm{i}\sin 47\theta)-(\cos(-\theta)+\mathrm{i}\sin(-\theta))+(\cos 45\theta+\mathrm{i}\sin 45\theta)}{2-2\cos 2\theta}$

$= \frac{(\cos\theta+\mathrm{i}\sin\theta)-(\cos 47\theta+\mathrm{i}\sin 47\theta)-(\cos\theta-\mathrm{i}\sin\theta)+(\cos 45\theta+\mathrm{i}\sin 45\theta)}{2-2\cos 2\theta}$

$= \frac{\cos\theta-\cos 47\theta-\cos\theta+\cos 45\theta+\mathrm{i}(\sin\theta-\sin 47\theta+\sin\theta+\sin 45\theta)}{2-2\cos 2\theta}$

Compare reals:

$C = \frac{-\cos 47\theta+\cos 45\theta}{2-2\cos 2\theta}$

as req.

16) c)

Compare imaginary:

$S = \frac{2\sin\theta-\sin 47\theta+\sin 45\theta}{2-2\cos 2\theta}$

as req.

17) a)

$\left(1-\frac{1}{9}\mathrm{e}^{\mathrm{i}3\theta}\right)\left(1-\frac{1}{9}\mathrm{e}^{-\mathrm{i}3\theta}\right) = 1-\frac{1}{9}\mathrm{e}^{\mathrm{i}3\theta}-\frac{1}{9}\mathrm{e}^{-\mathrm{i}3\theta}+\frac{1}{81}$

$= 1-\frac{1}{9}\left(\mathrm{e}^{\mathrm{i}3\theta}+\mathrm{e}^{-\mathrm{i}3\theta}\right)+\frac{1}{81}$

$= \frac{82}{81}-\frac{9}{81}(2\cos 3\theta)$

$= \frac{1}{81}(82-18\cos 3\theta)$ as req.

17) b)

$a = \mathrm{e}^{3\mathrm{i}\theta}$, $r = \frac{1}{9}\mathrm{e}^{\mathrm{i}3\theta}$

$S_\infty = \frac{a}{1-r} = \frac{\mathrm{e}^{\mathrm{i}3\theta}}{1-\frac{1}{9}\mathrm{e}^{\mathrm{i}3\theta}}$

17) c)

$C + \mathrm{i}S = \frac{\mathrm{e}^{\mathrm{i}3\theta}}{1-\frac{1}{9}\mathrm{e}^{\mathrm{i}3\theta}} \times \frac{1-\frac{1}{9}\mathrm{e}^{-\mathrm{i}3\theta}}{1-\frac{1}{9}\mathrm{e}^{-\mathrm{i}3\theta}} = \frac{\mathrm{e}^{\mathrm{i}3\theta}-\frac{1}{9}}{\frac{1}{81}(82-18\cos 3\theta)}$

$= \frac{\cos 3\theta+\mathrm{i}\sin 3\theta-\frac{1}{9}}{\frac{1}{81}(82-18\cos 3\theta)}$

$= \frac{81\cos 3\theta+81\mathrm{i}\sin 3\theta-9}{82-18\cos 3\theta}$

Compare reals:

$C = \frac{81\cos 3\theta-9}{82-18\cos 3\theta}$

17) d)

Compare imaginary:

$S = \frac{81\sin 3\theta}{82-18\cos 3\theta}$

18)

$C + \mathrm{i}S = \frac{1}{3}\cos\theta + \frac{1}{9}\cos 2\theta + \frac{1}{27}\cos 3\theta + \cdots + \frac{1}{3^n}\cos n\theta + \mathrm{i}\left(\frac{1}{3}\sin\theta + \frac{1}{9}\sin 2\theta + \frac{1}{27}\sin 3\theta + \cdots + \frac{1}{3^n}\sin n\theta\right)$

$= \frac{1}{3}(\cos\theta + \mathrm{i}\sin\theta) + \frac{1}{9}(\cos 2\theta + \mathrm{i}\sin 2\theta) + \cdots + \frac{1}{3^n}(\cos n\theta + \mathrm{i}\sin n\theta)$

$= \frac{1}{3}\mathrm{e}^{\mathrm{i}\theta} + \frac{1}{9}\mathrm{e}^{\mathrm{i}2\theta} + \frac{1}{27}\mathrm{e}^{\mathrm{i}3\theta} + \cdots + \frac{1}{3^n}\mathrm{e}^{\mathrm{i}n\theta}$

$a = \frac{1}{3}\mathrm{e}^{\mathrm{i}\theta}$, $r = \frac{1}{3}\mathrm{e}^{\mathrm{i}\theta}$, $n = n$

$S_n = \frac{a(1-r^n)}{1-r} = \frac{\frac{1}{3}\mathrm{e}^{\mathrm{i}\theta}\left(1-\frac{1}{3^n}\mathrm{e}^{\mathrm{i}n\theta}\right)}{1-\frac{1}{3}\mathrm{e}^{\mathrm{i}\theta}} = \frac{\mathrm{e}^{\mathrm{i}\theta}\left(1-\frac{1}{3^n}\mathrm{e}^{\mathrm{i}n\theta}\right)}{3-\mathrm{e}^{\mathrm{i}\theta}} = \frac{\mathrm{e}^{\mathrm{i}\theta}-\frac{1}{3^n}\mathrm{e}^{\mathrm{i}(n+1)\theta}}{3-\mathrm{e}^{\mathrm{i}\theta}}$

$C + \mathrm{i}S = \frac{\mathrm{e}^{\mathrm{i}\theta}-\frac{1}{3^n}\mathrm{e}^{\mathrm{i}(n+1)\theta}}{3-\mathrm{e}^{\mathrm{i}\theta}} \times \frac{3-\mathrm{e}^{-\mathrm{i}\theta}}{3-\mathrm{e}^{-\mathrm{i}\theta}} = \frac{3\mathrm{e}^{\mathrm{i}\theta}-1-\frac{3}{3^n}\mathrm{e}^{\mathrm{i}(n+1)\theta}+\frac{1}{3^n}\mathrm{e}^{\mathrm{i}n\theta}}{9-3\mathrm{e}^{\mathrm{i}\theta}-3\mathrm{e}^{-\mathrm{i}\theta}+1}$

$= \frac{3\mathrm{e}^{\mathrm{i}\theta}-1-3^{1-n}\mathrm{e}^{\mathrm{i}(n+1)\theta}+3^{-n}\mathrm{e}^{\mathrm{i}n\theta}}{10-3\left(\mathrm{e}^{\mathrm{i}\theta}+\mathrm{e}^{-\mathrm{i}\theta}\right)}$

$= \frac{3\mathrm{e}^{\mathrm{i}\theta}-1-3^{1-n}\mathrm{e}^{\mathrm{i}(n+1)\theta}+3^{-n}\mathrm{e}^{\mathrm{i}n\theta}}{10-6\cos\theta}$

$= \frac{3(\cos\theta+\mathrm{i}\sin\theta)-1-3^{1-n}\left(\cos\left((n+1)\theta\right)+\mathrm{i}\sin\left((n+1)\theta\right)\right)+3^{-n}(\cos(n\theta)+\mathrm{i}\sin(n\theta))}{10-6\cos\theta}$

$= \frac{3\cos\theta-1-3^{1-n}\cos\left((n+1)\theta\right)+3^{-n}\cos(n\theta)+\mathrm{i}\left(3\sin\theta-3^{1-n}\sin\left((n+1)\theta\right)+3^{-n}\sin(n\theta)\right)}{10-6\cos\theta}$

Compare reals:

$C = \frac{3\cos\theta-1-3^{1-n}\cos\left((n+1)\theta\right)+3^{-n}\cos(n\theta)}{10-6\cos\theta}$

$= \frac{3^n\left(3\cos\theta-1-3^{1-n}\cos\left((n+1)\theta\right)+3^{-n}\cos(n\theta)\right)}{3^n(10-6\cos\theta)}$

$= \frac{3^{n+1}\cos\theta-3^n-3\cos\left((n+1)\theta\right)+\cos(n\theta)}{3^n(10-6\cos\theta)}$

as req.

19) a)

$C + \mathrm{i}S = \cos\theta + \cos 3\theta + \cos 5\theta + \cdots + \cos(2n-1)\theta + \mathrm{i}(\sin\theta + \sin 3\theta + \sin 5\theta + \cdots + \sin(2n-1)\theta)$

$= (\cos\theta + \mathrm{i}\sin\theta) + (\cos 3\theta + \mathrm{i}\sin 3\theta) + \cdots + (\cos(2n-1)\theta + \mathrm{i}\sin(2n-1)\theta)$

$= \mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{\mathrm{i}3\theta} + \mathrm{e}^{\mathrm{i}5\theta} + \cdots + \mathrm{e}^{\mathrm{i}(2n-1)\theta}$

$a = \mathrm{e}^{\mathrm{i}\theta}$, $r = \mathrm{e}^{\mathrm{i}2\theta}$, $n = n$

$S_n = \frac{a(1-r^n)}{1-r} = \frac{\mathrm{e}^{\mathrm{i}\theta}(1-\mathrm{e}^{\mathrm{i}2n\theta})}{1-\mathrm{e}^{\mathrm{i}2\theta}} = \frac{\mathrm{e}^{\mathrm{i}\theta}-\mathrm{e}^{\mathrm{i}(2n+1)\theta}}{1-\mathrm{e}^{\mathrm{i}2\theta}}$

$C + \mathrm{i}S = \frac{\mathrm{e}^{\mathrm{i}\theta}-\mathrm{e}^{\mathrm{i}(2n+1)\theta}}{1-\mathrm{e}^{\mathrm{i}2\theta}} \times \frac{1-\mathrm{e}^{-\mathrm{i}2\theta}}{1-\mathrm{e}^{-\mathrm{i}2\theta}} = \frac{\mathrm{e}^{\mathrm{i}\theta}-\mathrm{e}^{\mathrm{i}(2n+1)\theta}-(\mathrm{e}^{-\mathrm{i}\theta}-\mathrm{e}^{\mathrm{i}(2n-1)\theta})}{1-\mathrm{e}^{\mathrm{i}2\theta}-\mathrm{e}^{-\mathrm{i}2\theta}+1}$

$= \frac{\mathrm{e}^{\mathrm{i}\theta}-\mathrm{e}^{\mathrm{i}(2n+1)\theta}-\mathrm{e}^{-\mathrm{i}\theta}+\mathrm{e}^{\mathrm{i}(2n-1)\theta}}{2-(\mathrm{e}^{\mathrm{i}2\theta}+\mathrm{e}^{-\mathrm{i}2\theta})}$

$= \frac{\mathrm{e}^{\mathrm{i}\theta}-\mathrm{e}^{\mathrm{i}(2n+1)\theta}-\mathrm{e}^{-\mathrm{i}\theta}+\mathrm{e}^{\mathrm{i}(2n-1)\theta}}{2-2\cos 2\theta}$

$= \frac{(\cos\theta+\mathrm{i}\sin\theta)-(\cos((2n+1)\theta)+\mathrm{i}\sin((2n+1)\theta))-(\cos(-\theta)+\mathrm{i}\sin(-\theta))+(\cos((2n-1)\theta)+\mathrm{i}\sin((2n-1)\theta))}{2-2\cos 2\theta}$

$= \frac{(\cos\theta+\mathrm{i}\sin\theta)-(\cos((2n+1)\theta)+\mathrm{i}\sin((2n+1)\theta))-(\cos\theta-\mathrm{i}\sin\theta)+(\cos((2n-1)\theta)+\mathrm{i}\sin((2n-1)\theta))}{2-2\cos 2\theta}$

$= \frac{\cos\theta-\cos((2n+1)\theta)-\cos\theta+\cos((2n-1)\theta)+\mathrm{i}(\sin\theta-\sin((2n+1)\theta)+\sin\theta+\sin((2n-1)\theta))}{2-2\cos 2\theta}$

Compare reals:

$C = \frac{-\cos((2n+1)\theta)+\cos((2n-1)\theta)}{2-2\cos 2\theta}$

as req.

19) b)

Compare imaginary:

$S = \frac{2\sin\theta-\sin((2n+1)\theta)+\sin((2n-1)\theta)}{2-2\cos 2\theta}$

as req.

20) a)

$(1+\mathrm{e}^{\mathrm{i}\theta})^4 = 1 + 4\mathrm{e}^{\mathrm{i}\theta} + 6\mathrm{e}^{\mathrm{i}2\theta} + 4\mathrm{e}^{\mathrm{i}3\theta} + \mathrm{e}^{\mathrm{i}4\theta}$

20) b)

$(1+\mathrm{e}^{\mathrm{i}\theta})^5 = 1 + 5\mathrm{e}^{\mathrm{i}\theta} + 10\mathrm{e}^{\mathrm{i}2\theta} + 10\mathrm{e}^{\mathrm{i}3\theta} + 5\mathrm{e}^{\mathrm{i}4\theta} + \mathrm{e}^{\mathrm{i}5\theta}$

20) c)

$$\left(1 + \mathrm{e}^{\mathrm{i}\theta}\right)^n = 1 + \binom{n}{1}\mathrm{e}^{\mathrm{i}\theta} + \binom{n}{2}\mathrm{e}^{\mathrm{i}2\theta} + \cdots + \binom{n}{n}\mathrm{e}^{\mathrm{i}n\theta}$$

20) d)

$$\left(1 + \mathrm{e}^{\mathrm{i}2\theta}\right)^n = 1 + \binom{n}{1}\mathrm{e}^{\mathrm{i}2\theta} + \binom{n}{2}\mathrm{e}^{\mathrm{i}4\theta} + \cdots + \binom{n}{n}\mathrm{e}^{\mathrm{i}2n\theta}$$

21) a)

$1 + \cos 2A = 1 + (2\cos^2 A - 1)$

$= 2\cos^2 A$

21) b)

$\sin 2A = 2\sin A \cos A$

21) c)

$1 + \mathrm{e}^{\mathrm{i}\theta} = 1 + \cos\theta + \mathrm{i}\sin\theta$

Since $1 + \cos\theta = 2\cos^2\left(\frac{\theta}{2}\right)$ then

$1 + \mathrm{e}^{\mathrm{i}\theta} = 2\cos^2\left(\frac{\theta}{2}\right) + \mathrm{i}\sin\theta$

Since $\sin\theta = 2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)$ then

$1 + \mathrm{e}^{\mathrm{i}\theta} = 2\cos^2\left(\frac{\theta}{2}\right) + \mathrm{i}\left(2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)\right)$

$= 2\cos\left(\frac{\theta}{2}\right)\left(\cos\left(\frac{\theta}{2}\right) + \mathrm{i}\sin\left(\frac{\theta}{2}\right)\right)$

$= 2\mathrm{e}^{\mathrm{i}\frac{\theta}{2}}\cos\left(\frac{\theta}{2}\right)$ as req.

22) a)

RHS: $2\mathrm{e}^{\mathrm{i}\frac{\theta}{2}}\left(\cos\frac{\theta}{2}\right) = \mathrm{e}^{\mathrm{i}\frac{\theta}{2}}\left(2\cos\frac{\theta}{2}\right) = \mathrm{e}^{\mathrm{i}\frac{\theta}{2}}\left(\mathrm{e}^{\mathrm{i}\frac{\theta}{2}} + \mathrm{e}^{-\mathrm{i}\frac{\theta}{2}}\right) = \mathrm{e}^{\mathrm{i}\theta} + \frac{\mathrm{e}^{\mathrm{i}\theta}}{\mathrm{e}^{\mathrm{i}\theta}} = \mathrm{e}^{\mathrm{i}\theta} + 1$:LHS ∴ True.

22) b)

$C + \mathrm{i}S = 1 + 4\cos\theta + 6\cos 2\theta + 4\cos 3\theta + \cos 4\theta + \mathrm{i}(4\sin\theta + 6\sin 2\theta + 4\sin 3\theta + \sin 4\theta)$

$= 1 + 4(\cos\theta + \mathrm{i}\sin\theta) + 6(\cos 2\theta + \mathrm{i}\sin 2\theta) + \cdots + (\cos 5\theta + \mathrm{i}\sin 5\theta)$

$= 1 + 4\mathrm{e}^{\mathrm{i}\theta} + 6\mathrm{e}^{\mathrm{i}2\theta} + 4\mathrm{e}^{\mathrm{i}3\theta} + \mathrm{e}^{\mathrm{i}4\theta}$

$= \left(1 + \mathrm{e}^{\mathrm{i}\theta}\right)^4$

Using $1 + \mathrm{e}^{\mathrm{i}\theta} = 2\mathrm{e}^{\mathrm{i}\frac{\theta}{2}}\cos\left(\frac{\theta}{2}\right)$ then:

$C + \mathrm{i}S = \left(2\mathrm{e}^{\mathrm{i}\frac{\theta}{2}}\cos\left(\frac{\theta}{2}\right)\right)^4$

$= 16\mathrm{e}^{\mathrm{i}\frac{4}{2}\theta}\cos^4\left(\frac{\theta}{2}\right)$

$= 16(\cos(2\theta) + \mathrm{i}\sin(2\theta))\cos^4\left(\frac{\theta}{2}\right)$

Compare reals:

$C = 16\cos(2\theta)\cos^4\left(\frac{\theta}{2}\right)$ as req.

22) c)

Compare imaginary:

$S = 16\sin(2\theta)\cos^4\left(\frac{\theta}{2}\right)$ as req.

23) a)

$1 - \mathrm{e}^{\mathrm{i}\theta} = 1 - \cos\theta - \mathrm{i}\sin\theta$

Since $1 - \cos\theta = 2\sin^2\left(\frac{\theta}{2}\right)$ then

$1 - \mathrm{e}^{\mathrm{i}\theta} = 2\sin^2\left(\frac{\theta}{2}\right) - \mathrm{i}\sin\theta$

Since $\sin\theta = 2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)$ then

$1 - \mathrm{e}^{\mathrm{i}\theta} = 2\sin^2\left(\frac{\theta}{2}\right) - \mathrm{i}\left(2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)\right)$

$= 2\sin\left(\frac{\theta}{2}\right)\left(\sin\left(\frac{\theta}{2}\right) - \mathrm{i}\cos\left(\frac{\theta}{2}\right)\right)$

$= 2\mathrm{i}\sin\left(\frac{\theta}{2}\right)\left(-\cos\left(\frac{\theta}{2}\right) + \frac{1}{\mathrm{i}}\sin\left(\frac{\theta}{2}\right)\right)$

$= -2\mathrm{i}\sin\left(\frac{\theta}{2}\right)\left(\cos\left(\frac{\theta}{2}\right) + \mathrm{i}\sin\left(\frac{\theta}{2}\right)\right)$

$= -2\mathrm{i}\mathrm{e}^{\mathrm{i}\frac{\theta}{2}}\sin\left(\frac{\theta}{2}\right)$ as req.

Alternatively:

RHS: $-2\mathrm{i}\mathrm{e}^{\mathrm{i}\frac{\theta}{2}}\left(\sin\frac{\theta}{2}\right) = -\mathrm{e}^{\mathrm{i}\frac{\theta}{2}}\left(2\mathrm{i}\sin\frac{\theta}{2}\right) = -\mathrm{e}^{\mathrm{i}\frac{\theta}{2}}\left(\mathrm{e}^{\mathrm{i}\frac{\theta}{2}} - \mathrm{e}^{-\mathrm{i}\frac{\theta}{2}}\right) = -\mathrm{e}^{\mathrm{i}\theta} + \frac{\mathrm{e}^{\mathrm{i}\frac{\theta}{2}}}{\mathrm{e}^{\mathrm{i}\frac{\theta}{2}}} = 1 - \mathrm{e}^{\mathrm{i}\theta}$

Which is the LHS ∴ True.

23) b)

$C + \mathrm{i}S = 1 - 5\cos\theta + 10\cos 2\theta + \cdots - \cos 5\theta + \mathrm{i}(-5\sin\theta + 10\sin 2\theta + \cdots - \sin 5\theta)$

$= 1 - 5(\cos\theta + \mathrm{i}\sin\theta) + 10(\cos 2\theta + \mathrm{i}\sin 2\theta) + \cdots - (\cos 5\theta + \mathrm{i}\sin 5\theta)$

$= 1 - 5\mathrm{e}^{\mathrm{i}\theta} + 10\mathrm{e}^{\mathrm{i}2\theta} - 10\mathrm{e}^{\mathrm{i}3\theta} + 5\mathrm{e}^{\mathrm{i}4\theta} - \mathrm{e}^{\mathrm{i}5\theta}$

$= \left(1 - \mathrm{e}^{\mathrm{i}\theta}\right)^5$

Using $1 - \mathrm{e}^{\mathrm{i}\theta} = -2\mathrm{i}\mathrm{e}^{\mathrm{i}\frac{\theta}{2}}\sin\left(\frac{\theta}{2}\right)$ then:

$C + \mathrm{i}S = \left(-2\mathrm{i}\mathrm{e}^{\mathrm{i}\frac{\theta}{2}}\sin\left(\frac{\theta}{2}\right)\right)^5$

$= -32\mathrm{i}\mathrm{e}^{\mathrm{i}\frac{5}{2}\theta}\sin^5\left(\frac{\theta}{2}\right)$

$= -32\mathrm{i}\left(\cos\left(\frac{5}{2}\theta\right) + \mathrm{i}\sin\left(\frac{5}{2}\theta\right)\right)\sin^5\left(\frac{\theta}{2}\right)$

$= \left(-32\mathrm{i}\cos\left(\frac{5}{2}\theta\right) + 32\sin\left(\frac{5}{2}\theta\right)\right)\sin^5\left(\frac{\theta}{2}\right)$

Compare reals:

$C = 32\sin\left(\frac{5}{2}\theta\right)\sin^5\left(\frac{\theta}{2}\right)$ as req.

23) c)

Compare imaginary:

$S = -32\cos\left(\frac{5}{2}\theta\right)\sin^5\left(\frac{\theta}{2}\right)$ as req.

24) a)

$C + \mathrm{i}S = 1 + 6\cos\theta + 15\cos 2\theta + \cdots + \cos 6\theta + \mathrm{i}(6\sin\theta + 15\sin 2\theta + \cdots + \sin 6\theta)$

$= 1 + 6(\cos\theta + \mathrm{i}\sin\theta) + 15(\cos 2\theta + \mathrm{i}\sin 2\theta) + \cdots + (\cos 6\theta + \mathrm{i}\sin 6\theta)$

$= 1 + 6\mathrm{e}^{\mathrm{i}\theta} + 15\mathrm{e}^{\mathrm{i}2\theta} + 20\mathrm{e}^{\mathrm{i}3\theta} + 15\mathrm{e}^{\mathrm{i}4\theta} + 6\mathrm{e}^{\mathrm{i}5\theta} + \mathrm{e}^{\mathrm{i}6\theta}$

$= \left(1 + \mathrm{e}^{\mathrm{i}\theta}\right)^6$

Using $1 + \mathrm{e}^{\mathrm{i}\theta} = 2\mathrm{e}^{\mathrm{i}\frac{\theta}{2}}\cos\left(\frac{\theta}{2}\right)$ then:

$C + \mathrm{i}S = \left(2\mathrm{e}^{\mathrm{i}\frac{\theta}{2}}\cos\left(\frac{\theta}{2}\right)\right)^6$

$= 64\mathrm{e}^{\mathrm{i}3\theta}\cos^6\left(\frac{\theta}{2}\right)$

$= 64(\cos 3\theta + \mathrm{i}\sin 3\theta)\cos^6\left(\frac{\theta}{2}\right)$

Compare reals:

$C = 64\cos 3\theta\cos^6\left(\frac{\theta}{2}\right)$ as req.

24) b)

Compare imaginary:

$S = 64\sin 3\theta\cos^6\left(\frac{\theta}{2}\right)$ as req.

25) a)

$C + \mathrm{i}S = 1 + 6\cos 2\theta + 15\cos 4\theta + \cdots + \cos 12\theta + \mathrm{i}(6\sin 2\theta + 15\sin 4\theta + \cdots + \sin 12\theta)$

$= 1 + 6(\cos 2\theta + \mathrm{i}\sin 2\theta) + 15(\cos 4\theta + \mathrm{i}\sin 4\theta) + \cdots + (\cos 12\theta + \mathrm{i}\sin 12\theta)$

$= 1 + 6\mathrm{e}^{\mathrm{i}2\theta} + 15\mathrm{e}^{\mathrm{i}4\theta} + 20\mathrm{e}^{\mathrm{i}6\theta} + 15\mathrm{e}^{\mathrm{i}8\theta} + 6\mathrm{e}^{\mathrm{i}10\theta} + \mathrm{e}^{\mathrm{i}12\theta}$

$= \left(1 + \mathrm{e}^{\mathrm{i}2\theta}\right)^6$

Using $1 + \mathrm{e}^{\mathrm{i}\theta} = 2\mathrm{e}^{\mathrm{i}\frac{\theta}{2}}\cos\left(\frac{\theta}{2}\right)$ then:

$C + \mathrm{i}S = \left(2\mathrm{e}^{\mathrm{i}\theta}\cos\theta\right)^6$

$= 64\mathrm{e}^{\mathrm{i}6\theta}\cos^6\theta$

$= 64(\cos 6\theta + \mathrm{i}\sin 6\theta)\cos^6\theta$

Compare reals:

$C = 64\cos 6\theta\cos^6\theta$ as req.

25) b)

Compare imaginary:

$S = 64\sin 6\theta\cos^6\theta$ as req.

26) a)

$C + \mathrm{i}S = 1 + \binom{n}{1}\cos\theta + \binom{n}{2}\cos 2\theta + \cdots + \binom{n}{n}\cos n\theta + \mathrm{i}\left(\binom{n}{1}\sin\theta + \binom{n}{2}\sin 2\theta + \cdots + \binom{n}{n}\sin n\theta\right)$

$= 1 + \binom{n}{1}(\cos\theta + \mathrm{i}\sin\theta) + \binom{n}{2}(\cos 2\theta + \mathrm{i}\sin 2\theta) + \cdots + \binom{n}{n}(\cos n\theta + \mathrm{i}\sin n\theta)$

$= 1 + \binom{n}{1}\mathrm{e}^{\mathrm{i}\theta} + \binom{n}{2}\mathrm{e}^{\mathrm{i}2\theta} + \cdots + \binom{n}{n}\mathrm{e}^{\mathrm{i}n\theta}$

$= \left(1 + \mathrm{e}^{\mathrm{i}\theta}\right)^n$

Using $1 + \mathrm{e}^{\mathrm{i}\theta} = 2\mathrm{e}^{\mathrm{i}\frac{\theta}{2}}\cos\left(\frac{\theta}{2}\right)$ then:

$C + \mathrm{i}S = \left(2\mathrm{e}^{\mathrm{i}\frac{\theta}{2}}\cos\left(\frac{\theta}{2}\right)\right)^n$

$= 2^n\mathrm{e}^{\mathrm{i}\frac{n}{2}\theta}\cos^n\left(\frac{\theta}{2}\right)$

$= 2^n\left(\cos\left(\frac{n}{2}\theta\right) + \mathrm{i}\sin\left(\frac{n}{2}\theta\right)\right)\cos^n\left(\frac{\theta}{2}\right)$

Compare reals:

$C = 2^n\cos\left(\frac{n}{2}\theta\right)\cos^n\left(\frac{\theta}{2}\right)$ as req.

26) b)

Compare imaginary:

$S = 2^n \sin\left(\frac{n}{2}\theta\right)\cos^n\left(\frac{\theta}{2}\right)$ as req.

27) a)

$1 + \cos 2A = 1 + (2\cos^2 A - 1) = 2\cos^2 A$ and $\sin 2A = 2\sin A \cos B$

Given that $1 + \mathrm{e}^{\mathrm{i}\theta} = 1 + \cos\theta + \mathrm{i}\sin\theta$

And since $1 + \cos\theta = 2\cos^2\left(\frac{\theta}{2}\right)$ then

$1 + \mathrm{e}^{\mathrm{i}\theta} = 2\cos^2\left(\frac{\theta}{2}\right) + \mathrm{i}\sin\theta$

Also since $\sin\theta = 2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)$ then

$1 + \mathrm{e}^{\mathrm{i}\theta} = 2\cos^2\left(\frac{\theta}{2}\right) + \mathrm{i}\left(2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)\right)$

$= 2\cos\left(\frac{\theta}{2}\right)\left(\cos\left(\frac{\theta}{2}\right) + \mathrm{i}\sin\left(\frac{\theta}{2}\right)\right)$

$1 + \mathrm{e}^{\mathrm{i}\theta} = 2\mathrm{e}^{\mathrm{i}\frac{\theta}{2}}\cos\left(\frac{\theta}{2}\right)$ as req.

27) b)

$C + \mathrm{i}S = 1 + 3\cos\theta + 3\cos 2\theta + \cos 3\theta + \mathrm{i}(3\sin\theta + 3\sin 2\theta + \sin 3\theta)$

$= 1 + 3(\cos\theta + \mathrm{i}\sin\theta) + 3(\cos 2\theta + \mathrm{i}\sin 2\theta) + (\cos 3\theta + \mathrm{i}\sin 3\theta)$

$= 1 + 3\mathrm{e}^{\mathrm{i}\theta} + 3\mathrm{e}^{\mathrm{i}2\theta} + \mathrm{e}^{\mathrm{i}3\theta}$

$= \left(1 + \mathrm{e}^{\mathrm{i}\theta}\right)^3$

Using $1 + \mathrm{e}^{\mathrm{i}\theta} = 2\mathrm{e}^{\mathrm{i}\frac{\theta}{2}}\cos\left(\frac{\theta}{2}\right)$ then:

$C + \mathrm{i}S = \left(2\mathrm{e}^{\mathrm{i}\frac{\theta}{2}}\cos\left(\frac{\theta}{2}\right)\right)^3$

$= 8\mathrm{e}^{\mathrm{i}\frac{3}{2}\theta}\cos^3\left(\frac{\theta}{2}\right)$

$= 8\left(\cos\left(\frac{3}{2}\theta\right) + \mathrm{i}\sin\left(\frac{3}{2}\theta\right)\right)\cos^3\left(\frac{\theta}{2}\right)$

Compare reals:

$C = 8\cos\left(\frac{3}{2}\theta\right)\cos^3\left(\frac{\theta}{2}\right)$ as req.

Compare imaginary:

$S = 8\sin\left(\frac{3}{2}\theta\right)\cos^3\left(\frac{\theta}{2}\right)$ as req.

$\therefore C = 8\cos\left(\frac{3}{2}\theta\right)\cos^3\left(\frac{\theta}{2}\right)$ and $S = 8\sin\left(\frac{3}{2}\theta\right)\cos^3\left(\frac{\theta}{2}\right)$

28) a)

$a = 1, r = z$

$S_\infty = \frac{a}{1-r} = \frac{1}{1-z}$

28) b)

If $z = \frac{1}{3}(\cos\theta + \mathrm{i}\sin\theta)$ then $z = \frac{1}{3}\mathrm{e}^{\mathrm{i}\theta}$

$\frac{1}{3}\sin\theta + \frac{1}{9}\sin 2\theta + \frac{1}{27}\sin 3\theta + \cdots$ is the imaginary part of series $\sum_{n=1}^{\infty}\frac{1}{3^n}\mathrm{e}^{\mathrm{i}n\theta}$

$a = \frac{1}{3}\mathrm{e}^{\mathrm{i}\theta}, r = \frac{1}{3}\mathrm{e}^{\mathrm{i}\theta}$

$S_\infty = \frac{a}{1-r} = \frac{\frac{1}{3}\mathrm{e}^{\mathrm{i}\theta}}{1-\frac{1}{3}\mathrm{e}^{\mathrm{i}\theta}} = \frac{\mathrm{e}^{\mathrm{i}\theta}}{3-\mathrm{e}^{\mathrm{i}\theta}}$

$= \frac{\mathrm{e}^{\mathrm{i}\theta}}{3-\mathrm{e}^{\mathrm{i}\theta}} \times \frac{3-\mathrm{e}^{-\mathrm{i}\theta}}{3-\mathrm{e}^{-\mathrm{i}\theta}} = \frac{3\mathrm{e}^{\mathrm{i}\theta}-1}{9-3\left(\mathrm{e}^{\mathrm{i}\theta}+\mathrm{e}^{-\mathrm{i}\theta}\right)+1} = \frac{3(\cos\theta+\mathrm{i}\sin\theta)-1}{10-6\cos\theta}$

$= \frac{3\cos\theta-1+\mathrm{i}(3\sin\theta)}{10-6\cos\theta}$

Compare imaginary:

$\frac{1}{3}\sin\theta + \frac{1}{9}\sin 2\theta + \frac{1}{27}\sin 3\theta + \cdots = \frac{3\sin\theta}{10-6\cos\theta}$

28) c)

$$\sum_{n=0}^{\infty} z^n = \frac{1}{1-z}$$

For a purely imaginary number, $\mathrm{Re}\left(\frac{1}{1-z}\right) = 0$

But $|z| < 1$ and so $\mathrm{Re}(1-z) > 0 \therefore \mathrm{Re}\left(\frac{1}{1-z}\right) \neq 0$

29) a)

$a = z, r = z$

$S_\infty = \frac{a}{1-r} = \frac{z}{1-z}$

29) b)

If $z = \frac{1}{4}(\cos\theta + \mathrm{i}\sin\theta)$ then $z = \frac{1}{4}\mathrm{e}^{\mathrm{i}\theta}$

$\frac{1}{4}\cos\theta + \frac{1}{16}\cos 2\theta + \frac{1}{64}\cos 3\theta + \cdots$ is the real part of series $\sum_{n=1}^{\infty}\frac{1}{4^n}\mathrm{e}^{\mathrm{i}n\theta}$

$a = \frac{1}{4}\mathrm{e}^{\mathrm{i}\theta}$, $r = \frac{1}{4}\mathrm{e}^{\mathrm{i}\theta}$

$S_\infty = \frac{a}{1-r} = \frac{\frac{1}{4}\mathrm{e}^{\mathrm{i}\theta}}{1-\frac{1}{4}\mathrm{e}^{\mathrm{i}\theta}} = \frac{\mathrm{e}^{\mathrm{i}\theta}}{4-\mathrm{e}^{\mathrm{i}\theta}}$

$= \frac{\mathrm{e}^{\mathrm{i}\theta}}{4-\mathrm{e}^{\mathrm{i}\theta}} \times \frac{4-\mathrm{e}^{-\mathrm{i}\theta}}{4-\mathrm{e}^{-\mathrm{i}\theta}} = \frac{4\mathrm{e}^{\mathrm{i}\theta}-1}{16-4\left(\mathrm{e}^{\mathrm{i}\theta}+\mathrm{e}^{-\mathrm{i}\theta}\right)+1} = \frac{4(\cos\theta+\mathrm{i}\sin\theta)-1}{17-8\cos\theta}$

$= \frac{4\cos\theta-1+\mathrm{i}(4\sin\theta)}{17-8\cos\theta}$

Compare real:

$\frac{1}{4}\cos\theta + \frac{1}{16}\cos 2\theta + \frac{1}{64}\cos 3\theta + \cdots = \frac{4\cos\theta-1}{17-8\cos\theta}$

29) c)

Imaginary part:

$\frac{4\sin\theta}{17-8\cos\theta}$

This must be zero for it to be purely real, so $4\sin\theta = 0$

$\therefore \theta = k\pi$ where $k \in \mathbb{Z}$

30) a)

Since $z = \frac{1}{2}\mathrm{e}^{\mathrm{i}\theta}$

$\frac{|z^{n+1}|}{|z^n|} = \frac{\left|\frac{1}{2^{n+1}}\mathrm{e}^{\mathrm{i}(n+1)\theta}\right|}{\left|\frac{1}{2^n}\mathrm{e}^{\mathrm{i}n\theta}\right|} = \left|\frac{\frac{1}{2^{n+1}}\mathrm{e}^{\mathrm{i}(n+1)\theta}}{\frac{1}{2^n}\mathrm{e}^{\mathrm{i}n\theta}}\right| = \left|\frac{\frac{2^n}{2^{n+1}}\mathrm{e}^{\mathrm{i}n\theta+\mathrm{i}\theta}}{\mathrm{e}^{\mathrm{i}n\theta}}\right| = \frac{1}{2}\left|\mathrm{e}^{\mathrm{i}\theta}\right| = \frac{1}{2}$

30) b)

θ

30) c)

$\frac{1}{2}\sin\theta + \frac{1}{4}\sin 2\theta + \frac{1}{8}\sin 3\theta + \cdots$ is the imaginary part of series $\sum_{n=1}^{\infty} \frac{1}{2^n} \mathrm{e}^{\mathrm{i}n\theta}$

$a = \frac{1}{2}\mathrm{e}^{\mathrm{i}\theta}$, $r = \frac{1}{2}\mathrm{e}^{\mathrm{i}\theta}$

$S_\infty = \frac{a}{1-r} = \frac{\frac{1}{2}\mathrm{e}^{\mathrm{i}\theta}}{1-\frac{1}{2}\mathrm{e}^{\mathrm{i}\theta}} = \frac{\mathrm{e}^{\mathrm{i}\theta}}{2-\mathrm{e}^{\mathrm{i}\theta}}$

$= \frac{\mathrm{e}^{\mathrm{i}\theta}}{2-\mathrm{e}^{\mathrm{i}\theta}} \times \frac{2-\mathrm{e}^{-\mathrm{i}\theta}}{2-\mathrm{e}^{-\mathrm{i}\theta}} = \frac{2\mathrm{e}^{\mathrm{i}\theta}-1}{4-2\left(\mathrm{e}^{\mathrm{i}\theta}+\mathrm{e}^{-\mathrm{i}\theta}\right)+1} = \frac{2(\cos\theta+\mathrm{i}\sin\theta)-1}{5-4\cos\theta}$

$= \frac{2\cos\theta-1+\mathrm{i}(2\sin\theta)}{5-4\cos\theta}$

Compare imaginary:

$\frac{1}{2}\sin\theta + \frac{1}{4}\sin 2\theta + \frac{1}{8}\sin 3\theta + \cdots = \frac{2\sin\theta}{5-4\cos\theta}$

as req.

31)

$C + \mathrm{i}S = \frac{1}{2}\cos\theta + \frac{1}{4}\cos 2\theta + \frac{1}{8}\cos 3\theta + \cdots + \frac{1}{2^n}\cos n\theta + \mathrm{i}\left(\frac{1}{2}\sin\theta + \frac{1}{4}\sin 2\theta + \frac{1}{8}\sin 3\theta + \cdots + \frac{1}{2^n}\sin n\theta\right)$

$= \frac{1}{2}(\cos\theta + \mathrm{i}\sin\theta) + \frac{1}{4}(\cos 2\theta + \mathrm{i}\sin 2\theta) + \cdots + \frac{1}{2^n}(\cos n\theta + \mathrm{i}\sin n\theta)$

$= \frac{1}{2}\mathrm{e}^{\mathrm{i}\theta} + \frac{1}{4}\mathrm{e}^{\mathrm{i}2\theta} + \frac{1}{8}\mathrm{e}^{\mathrm{i}3\theta} + \cdots + \frac{1}{2^n}\mathrm{e}^{\mathrm{i}n\theta}$

$a = \frac{1}{2}\mathrm{e}^{\mathrm{i}\theta}$, $r = \frac{1}{2}\mathrm{e}^{\mathrm{i}\theta}$, $n = n$

$S_n = \frac{a(1-r^n)}{1-r} = \frac{\frac{1}{2}\mathrm{e}^{\mathrm{i}\theta}\left(1-\frac{1}{2^n}\mathrm{e}^{\mathrm{i}n\theta}\right)}{1-\frac{1}{2}\mathrm{e}^{\mathrm{i}\theta}} = \frac{\mathrm{e}^{\mathrm{i}\theta}\left(1-\frac{1}{2^n}\mathrm{e}^{\mathrm{i}n\theta}\right)}{2-\mathrm{e}^{\mathrm{i}\theta}} = \frac{\mathrm{e}^{\mathrm{i}\theta}-\frac{1}{2^n}\mathrm{e}^{\mathrm{i}(n+1)\theta}}{2-\mathrm{e}^{\mathrm{i}\theta}}$

$C + \mathrm{i}S = \frac{\mathrm{e}^{\mathrm{i}\theta}-\frac{1}{2^n}\mathrm{e}^{\mathrm{i}(n+1)\theta}}{2-\mathrm{e}^{\mathrm{i}\theta}} \times \frac{2-\mathrm{e}^{-\mathrm{i}\theta}}{2-\mathrm{e}^{-\mathrm{i}\theta}} = \frac{2\mathrm{e}^{\mathrm{i}\theta}-1-\frac{2}{2^n}\mathrm{e}^{\mathrm{i}(n+1)\theta}+\frac{1}{2^n}\mathrm{e}^{\mathrm{i}n\theta}}{4-2\mathrm{e}^{\mathrm{i}\theta}-2\mathrm{e}^{-\mathrm{i}\theta}+1}$

$= \frac{2\mathrm{e}^{\mathrm{i}\theta}-1-2^{1-n}\mathrm{e}^{\mathrm{i}(n+1)\theta}+2^{-n}\mathrm{e}^{\mathrm{i}n\theta}}{5-2\left(\mathrm{e}^{\mathrm{i}\theta}+\mathrm{e}^{-\mathrm{i}\theta}\right)}$

$= \frac{2\mathrm{e}^{\mathrm{i}\theta}-1-2^{1-n}\mathrm{e}^{\mathrm{i}(n+1)\theta}+2^{-n}\mathrm{e}^{\mathrm{i}n\theta}}{5-4\cos\theta}$

$= \frac{2(\cos\theta+\mathrm{i}\sin\theta)-1-2^{1-n}\left(\cos\left((n+1)\theta\right)+\mathrm{i}\sin\left((n+1)\theta\right)\right)+2^{-n}(\cos(n\theta)+\mathrm{i}\sin(n\theta))}{5-4\cos\theta}$

$= \frac{2\cos\theta-1-2^{1-n}\cos\left((n+1)\theta\right)+2^{-n}\cos(n\theta)+\mathrm{i}\left(2\sin\theta-2^{1-n}\sin\left((n+1)\theta\right)+2^{-n}\sin(n\theta)\right)}{5-4\cos\theta}$

Compare imaginary:

$S = \frac{2\sin\theta-2^{1-n}\sin\left((n+1)\theta\right)+2^{-n}\sin(n\theta)}{5-4\cos\theta}$

$= \frac{2^n\left(2\sin\theta-2^{1-n}\sin\left((n+1)\theta\right)+2^{-n}\sin(n\theta)\right)}{2^n(5-4\cos\theta)}$

$= \frac{2^{n+1}\sin\theta-2\sin\left((n+1)\theta\right)+\sin(n\theta)}{2^n(5-4\cos\theta)}$

as req.

32)

$C + \mathrm{i}S = 1 + 12\cos\theta + \cdots + \cos 12\theta + \mathrm{i}(1 + 12\sin\theta + \cdots + \sin 12\theta)$

$= 1 + 12(\cos\theta + \mathrm{i}\sin\theta) + \cdots + (\cos 12\theta + \mathrm{i}\sin 12\theta)$

$= 1 + 12\mathrm{e}^{\mathrm{i}\theta} + \cdots + \mathrm{e}^{\mathrm{i}12\theta}$

$= \left(1 + \mathrm{e}^{\mathrm{i}\theta}\right)^{12}$

Using $1 + \mathrm{e}^{\mathrm{i}\theta} = 2\mathrm{e}^{\mathrm{i}\frac{\theta}{2}}\cos\left(\frac{\theta}{2}\right)$ then:

$C + \mathrm{i}S = \left(2\mathrm{e}^{\mathrm{i}\frac{\theta}{2}}\cos\left(\frac{\theta}{2}\right)\right)^{12}$

$= 4096\mathrm{e}^{\mathrm{i}\frac{12}{2}\theta}\cos^{12}\left(\frac{\theta}{2}\right)$

$= 4096(\cos(6\theta) + \mathrm{i}\sin(6\theta))\cos^{12}\left(\frac{\theta}{2}\right)$

Compare imaginary:

$\therefore S = 4096\sin(6\theta)\cos^{12}\left(\frac{\theta}{2}\right)$ as req.

33)

$C + \mathrm{i}S = 1 - 16\cos\theta + \cdots + \cos 16\theta + \mathrm{i}(1 - 16\sin\theta + \cdots + \sin 16\theta)$

$= 1 - 16(\cos\theta + \mathrm{i}\sin\theta) + \cdots + (\cos 16\theta + \mathrm{i}\sin 16\theta)$

$= 1 - 16\mathrm{e}^{\mathrm{i}\theta} + \cdots + \mathrm{e}^{\mathrm{i}16\theta}$

$= \left(1 - \mathrm{e}^{\mathrm{i}\theta}\right)^{16}$

Using $1 - \mathrm{e}^{\mathrm{i}\theta} = -2\mathrm{i}\mathrm{e}^{\mathrm{i}\frac{\theta}{2}}\sin\left(\frac{\theta}{2}\right)$ then:

$C + \mathrm{i}S = \left(-2\mathrm{i}\mathrm{e}^{\mathrm{i}\frac{\theta}{2}}\sin\left(\frac{\theta}{2}\right)\right)^{16}$

$= 2^{16}\mathrm{e}^{\mathrm{i}\frac{16}{2}\theta}\sin^{16}\left(\frac{\theta}{2}\right)$

$= 2^{16}(\cos(8\theta) + \mathrm{i}\sin(8\theta))\sin^{16}\left(\frac{\theta}{2}\right)$

Compare real:

$\therefore C = 2^{16}\cos(8\theta)\sin^{16}\left(\frac{\theta}{2}\right)$ as req.

34) a)

$\frac{2}{1-\mathrm{e}^{\mathrm{i}2\theta}} \times \frac{1-\mathrm{e}^{-\mathrm{i}2\theta}}{1-\mathrm{e}^{-\mathrm{i}2\theta}} = \frac{2-2\mathrm{e}^{-\mathrm{i}2\theta}}{1-\mathrm{e}^{\mathrm{i}2\theta}-\mathrm{e}^{-\mathrm{i}2\theta}+1} = \frac{2-2\mathrm{e}^{-\mathrm{i}2\theta}}{2-\left(\mathrm{e}^{\mathrm{i}2\theta}+\mathrm{e}^{-\mathrm{i}2\theta}\right)}$

$= \frac{2-2\mathrm{e}^{-\mathrm{i}2\theta}}{2-2\cos 2\theta} = \frac{1-\mathrm{e}^{-\mathrm{i}2\theta}}{1-\cos 2\theta}$

Since $\mathrm{e}^{-\mathrm{i}2\theta} = \cos(-2\theta) + \mathrm{i}\sin(-2\theta) = \cos(2\theta) - \mathrm{i}\sin(2\theta)$ then

$1 - \mathrm{e}^{-\mathrm{i}2\theta} = 1 - \cos(2\theta) + \mathrm{i}\sin(2\theta)$

So

$\frac{1-\mathrm{e}^{-\mathrm{i}2\theta}}{2-2\cos 2\theta} = \frac{1-\cos(2\theta)+\mathrm{i}\sin(2\theta)}{1-\cos 2\theta} = 1 + \frac{\sin(2\theta)}{1-\cos 2\theta}\mathrm{i}$

$= 1 + \frac{2\cos\theta\sin\theta}{2\sin^2\theta}\mathrm{i}$

$= 1 + \frac{\cos\theta}{\sin\theta}\mathrm{i}$

$= 1 + \cot\theta\,\mathrm{i}$ as req.

34) b)

$C + \mathrm{i}S = 2 + 2\cos\frac{\pi}{14} + 2\cos\frac{2\pi}{14} + 2\cos\frac{3\pi}{14} + \cdots + 2\cos\frac{6\pi}{14} + 2\mathrm{i}\left(\sin\frac{\pi}{14} + \sin\frac{2\pi}{14} + \sin\frac{3\pi}{14} + \cdots + \sin\frac{6\pi}{14}\right)$

$= 2(\cos 0 + \mathrm{i}\sin 0) + 2\left(\cos\frac{\pi}{14} + \mathrm{i}\sin\frac{\pi}{14}\right) + 2\left(\cos\frac{2\pi}{14} + \mathrm{i}\sin\frac{2\pi}{14}\right) + \cdots + 2\left(\cos\frac{6\pi}{14} + \mathrm{i}\sin\frac{6\pi}{14}\right)$

$= 2\mathrm{e}^{0\mathrm{i}} + 2\mathrm{e}^{\frac{\pi}{14}\mathrm{i}} + 2\mathrm{e}^{\frac{2\pi}{14}\mathrm{i}} + 2\mathrm{e}^{\frac{3\pi}{14}\mathrm{i}} + \cdots + 2\mathrm{e}^{\frac{6\pi}{14}\mathrm{i}}$

$a = 2\mathrm{e}^{0\mathrm{i}}$, $r = \mathrm{e}^{\frac{\pi}{14}\mathrm{i}}$, $n = 7$

$S_n = \frac{a(1-r^n)}{1-r} = \frac{2\mathrm{e}^{0\mathrm{i}}\left(1-\mathrm{e}^{\frac{7\pi}{14}\mathrm{i}}\right)}{1-\mathrm{e}^{\frac{\pi}{14}\mathrm{i}}} = \frac{2\left(1-\mathrm{e}^{\frac{\pi}{2}\mathrm{i}}\right)}{1-\mathrm{e}^{\frac{\pi}{14}\mathrm{i}}}$

$C + \mathrm{i}S = \frac{2\left(1-\mathrm{e}^{\frac{\pi}{2}\mathrm{i}}\right)}{1-\mathrm{e}^{\frac{\pi}{14}\mathrm{i}}}$

But since $\frac{2}{1-\mathrm{e}^{\mathrm{i}2\theta}} = 1 + \cot\theta\,\mathrm{i}$ then set $\theta = \frac{\pi}{28}$:

$\frac{2}{1-\mathrm{e}^{\frac{\pi}{14}\mathrm{i}}} = 1 + \cot\frac{\pi}{28}\mathrm{i}$

So

$C + \mathrm{i}S = \frac{2\left(1-\mathrm{e}^{\frac{\pi}{2}\mathrm{i}}\right)}{1-\mathrm{e}^{\frac{\pi}{14}\mathrm{i}}} = \left(1 + \cot\frac{\pi}{28}\mathrm{i}\right)\left(1 - \mathrm{e}^{\frac{\pi}{2}\mathrm{i}}\right)$

$= \left(1 + \cot\frac{\pi}{28}\mathrm{i}\right)\left(1 - \left(\cos\frac{\pi}{2} + \mathrm{i}\sin\frac{\pi}{2}\right)\right)$

$= \left(1 + \cot\frac{\pi}{28}\mathrm{i}\right)(1 - \mathrm{i})$

$= 1 + \cot\frac{\pi}{28}\mathrm{i} - \mathrm{i} + \cot\frac{\pi}{28}$

So that

$C = 1 + \cot\frac{\pi}{28}$

34) c)

$S = -1 + \cot\frac{\pi}{28}$

And so

$C - 2 = 1 + \cot\frac{\pi}{28} - 2 = -1 + \cot\frac{\pi}{28} = S$

$\therefore S = C - 2$

35) a)

$C + \mathrm{i}S = \mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{\mathrm{i}\left(\theta+\frac{4\pi}{5}\right)} + \mathrm{e}^{\mathrm{i}\left(\theta+\frac{8\pi}{5}\right)} + \mathrm{e}^{\mathrm{i}\left(\theta+\frac{12\pi}{5}\right)} + \mathrm{e}^{\mathrm{i}\left(\theta+\frac{16\pi}{5}\right)}$

$= \mathrm{e}^{\mathrm{i}\theta}\left(1 + \mathrm{e}^{\frac{4\pi}{5}\mathrm{i}} + \mathrm{e}^{\frac{8\pi}{5}\mathrm{i}} + \mathrm{e}^{\frac{12\pi}{5}\mathrm{i}} + \mathrm{e}^{\frac{16\pi}{5}\mathrm{i}}\right)$

$= \mathrm{e}^{\mathrm{i}\theta}\sum_{k=0}^{4}\left(\mathrm{e}^{\frac{4\pi}{5}\mathrm{i}}\right)^{k}$

As req.

35) b)

$1 + \mathrm{e}^{\frac{4\pi}{5}\mathrm{i}} + \mathrm{e}^{\frac{8\pi}{5}\mathrm{i}} + \mathrm{e}^{\frac{12\pi}{5}\mathrm{i}} + \mathrm{e}^{\frac{16\pi}{5}\mathrm{i}}$ is a finite geometric series.

$a = 1$, $r = \mathrm{e}^{\frac{4\pi}{5}\mathrm{i}}$ and $n = 5$

$s_n = \frac{a(1-r^n)}{1-r} = \frac{1-\mathrm{e}^{4\pi\mathrm{i}}}{1-\mathrm{e}^{\frac{4\pi}{5}\mathrm{i}}} = \frac{1-1}{1-\mathrm{e}^{\frac{4\pi}{5}\mathrm{i}}} = 0$

$\therefore C + \mathrm{i}S = \mathrm{e}^{\mathrm{i}\theta} \times 0$

And so $C = 0$ and $S = 0$

36) a)

$C + \mathrm{i}S = \mathrm{e}^{0\mathrm{i}} - \mathrm{e}^{\frac{\pi}{2}\mathrm{i}} + \mathrm{e}^{\pi\mathrm{i}} - \mathrm{e}^{\frac{3\pi}{2}\mathrm{i}}$

$= \sum_{k=0}^{3}(-1)^k\left(\mathrm{e}^{\frac{\pi}{2}\mathrm{i}}\right)^{k}$

36) b)

$\mathrm{e}^{0\mathrm{i}} - \mathrm{e}^{\frac{\pi}{2}\mathrm{i}} + \mathrm{e}^{\pi\mathrm{i}} - \mathrm{e}^{\frac{3\pi}{2}\mathrm{i}}$ is a finite geometric series.

$a = 1$, $r = -\mathrm{e}^{\frac{\pi}{2}\mathrm{i}}$ and $n = 4$

$s_n = \frac{a(1-r^n)}{1-r} = \frac{1-\mathrm{e}^{2\pi\mathrm{i}}}{1+\mathrm{e}^{\frac{\pi}{2}\mathrm{i}}} = \frac{1-1}{1+\mathrm{e}^{\frac{\pi}{2}\mathrm{i}}} = 0$

$\therefore C + \mathrm{i}S = 0$

And so $C = 0$ and $S = 0$

37) a)

$C + \mathrm{i}S = 1 + k(\cos\theta + \mathrm{i}\sin\theta) + k^2(\cos 2\theta + \mathrm{i}\sin 2\theta) + k^3(\cos 3\theta + \mathrm{i}\sin 3\theta)\ldots$

$= 1 + k\mathrm{e}^{\mathrm{i}\theta} + k^2\mathrm{e}^{\mathrm{i}2\theta} + k^3\mathrm{e}^{\mathrm{i}3\theta} + \cdots$

$a = 1,\ r = k\mathrm{e}^{\mathrm{i}\theta}$

$S_\infty = \frac{a}{1-r} = \frac{1}{1-k\mathrm{e}^{\mathrm{i}\theta}}$

$C + \mathrm{i}S = \frac{1}{1-k\mathrm{e}^{\mathrm{i}\theta}} \times \frac{1-k\mathrm{e}^{-\mathrm{i}\theta}}{1-k\mathrm{e}^{-\mathrm{i}\theta}} = \frac{1-k\mathrm{e}^{-\mathrm{i}\theta}}{1-k\mathrm{e}^{\mathrm{i}\theta}-k\mathrm{e}^{-\mathrm{i}\theta}+k^2}$

$= \frac{1-k\mathrm{e}^{-\mathrm{i}\theta}}{1+k^2-k\left(\mathrm{e}^{\mathrm{i}\theta}+\mathrm{e}^{-\mathrm{i}\theta}\right)}$

$= \frac{1-k(\cos(-\theta)+i\sin(-\theta))}{1+k^2-2k\cos\theta}$

$= \frac{1-k\cos\theta+k\mathrm{i}\sin\theta}{1-2k\cos\theta+k^2}$

Compare reals:

$C = \frac{1-k\cos\theta}{1-2k\cos\theta+k^2}$ as req.

37) b)

Compare imaginary:

$S = \frac{k\sin\theta}{1-2k\cos\theta+k^2}$

38) a)

$C + \mathrm{i}S = k(\cos\theta + \mathrm{i}\sin\theta) + k^2(\cos 2\theta + \mathrm{i}\sin 2\theta) + k^3(\cos 3\theta + \mathrm{i}\sin 3\theta)\ldots$

$= k\mathrm{e}^{\mathrm{i}\theta} + k^2\mathrm{e}^{\mathrm{i}2\theta} + k^3\mathrm{e}^{\mathrm{i}3\theta} + \cdots$

$a = k\mathrm{e}^{\mathrm{i}\theta},\ r = k\mathrm{e}^{\mathrm{i}\theta}$

$S_\infty = \frac{a}{1-r} = \frac{k\mathrm{e}^{\mathrm{i}\theta}}{1-k\mathrm{e}^{\mathrm{i}\theta}}$

$C + \mathrm{i}S = \frac{k\mathrm{e}^{\mathrm{i}\theta}}{1-k\mathrm{e}^{\mathrm{i}\theta}} \times \frac{1-k\mathrm{e}^{-\mathrm{i}\theta}}{1-k\mathrm{e}^{-\mathrm{i}\theta}} = \frac{k\mathrm{e}^{\mathrm{i}\theta}-k^2}{1-k\mathrm{e}^{\mathrm{i}\theta}-k\mathrm{e}^{-\mathrm{i}\theta}+k^2}$

$= \frac{k\mathrm{e}^{\mathrm{i}\theta}-k^2}{1+k^2-k\left(\mathrm{e}^{\mathrm{i}\theta}+\mathrm{e}^{-\mathrm{i}\theta}\right)}$

$= \frac{k(\cos\theta+\mathrm{i}\sin\theta)-k^2}{1+k^2-2k\cos\theta}$

$= \frac{k\cos\theta+k\mathrm{i}\sin\theta-k^2}{1-2k\cos\theta+k^2}$

Compare reals:

$C = \frac{k\cos\theta-k^2}{1-2k\cos\theta+k^2}$ as req.

38) b)

Compare imaginary:

$S = \frac{k \sin\theta}{1 - 2k\cos\theta + k^2}$

39) a)

$C + \mathrm{i}S = 1 + \binom{n}{1}\cos\theta + \binom{n}{2}\cos 2\theta + \cdots + \binom{n}{n}\cos n\theta + \mathrm{i}\left(\binom{n}{1}\sin\theta + \binom{n}{2}\sin 2\theta + \cdots + \binom{n}{n}\sin n\theta\right)$

$= 1 + \binom{n}{1}(\cos 2\theta + \mathrm{i}\sin 2\theta) + \cdots + \binom{n}{n}(\cos n\theta + \mathrm{i}\sin n\theta)$

$= 1 + \binom{n}{1}\mathrm{e}^{\mathrm{i}2\theta} + \cdots + \binom{n}{n}\mathrm{e}^{\mathrm{i}n\theta}$

$= \left(1 + \mathrm{e}^{\mathrm{i}\theta}\right)^n$

Using $1 + \mathrm{e}^{\mathrm{i}\theta} = 2\mathrm{e}^{\mathrm{i}\frac{\theta}{2}}\cos\left(\frac{\theta}{2}\right)$ then:

$C + \mathrm{i}S = \left(2\mathrm{e}^{\mathrm{i}\frac{\theta}{2}}\cos\left(\frac{\theta}{2}\right)\right)^n$

$= 2^n\mathrm{e}^{\mathrm{i}\frac{n\theta}{2}}\cos^n\left(\frac{\theta}{2}\right)$

$= 2^n\left(\cos\left(\frac{n\theta}{2}\right) + \mathrm{i}\sin\left(\frac{n\theta}{2}\right)\right)\cos^n\left(\frac{\theta}{2}\right)$

Compare reals:

$C = 2^n\cos\left(\frac{n\theta}{2}\right)\cos^n\left(\frac{\theta}{2}\right)$ as req.

39) b)

Compare imaginary:

$S = 2^n\sin\left(\frac{n\theta}{2}\right)\cos^n\left(\frac{\theta}{2}\right)$

39) c)

$\frac{C}{S} = \frac{2^n\cos\left(\frac{n\theta}{2}\right)\cos^n\left(\frac{\theta}{2}\right)}{2^n\sin\left(\frac{n\theta}{2}\right)\cos^n\left(\frac{\theta}{2}\right)} = \frac{\cos\left(\frac{n\theta}{2}\right)}{\sin\left(\frac{n\theta}{2}\right)} = \cot\left(\frac{n\theta}{2}\right)$

40) a)

$1 - \mathrm{e}^{\mathrm{i}\theta} = 1 - \mathrm{e}^{\mathrm{i}\frac{\theta}{2}}\mathrm{e}^{\mathrm{i}\frac{\theta}{2}} = \mathrm{e}^{\mathrm{i}\frac{\theta}{2}}\left(\mathrm{e}^{-\mathrm{i}\frac{\theta}{2}} - \mathrm{e}^{\mathrm{i}\frac{\theta}{2}}\right) = -\mathrm{e}^{\mathrm{i}\frac{\theta}{2}}\left(\mathrm{e}^{\mathrm{i}\frac{\theta}{2}} - \mathrm{e}^{-\mathrm{i}\frac{\theta}{2}}\right)$

$= -\mathrm{e}^{\mathrm{i}\frac{\theta}{2}}\left(2\mathrm{i}\sin\frac{\theta}{2}\right)$

$= -2\mathrm{i}\mathrm{e}^{\mathrm{i}\frac{\theta}{2}}\sin\left(\frac{1}{2}\theta\right)$ as req.

40) b)

$C + \mathrm{i}S = 1 - \binom{n}{1}\cos\theta + \binom{n}{2}\cos 2\theta + \cdots + (-1)^n\binom{n}{n}\cos n\theta + \mathrm{i}\left(-\binom{n}{1}\sin\theta + \binom{n}{2}\sin 2\theta + \cdots + (-1)^n\binom{n}{n}\sin n\theta\right)$

$= 1 - \binom{n}{1}(\cos 2\theta + \mathrm{i}\sin 2\theta) + \cdots + (-1)^n\binom{n}{n}(\cos n\theta + \mathrm{i}\sin n\theta)$

$= 1 - \binom{n}{1}\mathrm{e}^{\mathrm{i}2\theta} + \cdots + (-1)^n\binom{n}{n}\mathrm{e}^{\mathrm{i}n\theta}$

$= \left(1 - \mathrm{e}^{\mathrm{i}\theta}\right)^n$

Using $1 - \mathrm{e}^{\mathrm{i}\theta} = -2\mathrm{i}\mathrm{e}^{\mathrm{i}\frac{\theta}{2}}\sin\frac{\theta}{2}$ then:

$C + \mathrm{i}S = \left(-2\mathrm{i}\mathrm{e}^{\mathrm{i}\frac{\theta}{2}}\sin\frac{\theta}{2}\right)^n$

$= \left(-2\mathrm{i}\left(\cos\frac{\theta}{2} + \mathrm{i}\sin\frac{\theta}{2}\right)\sin\frac{\theta}{2}\right)^n$ as req.

40) c)

Let $n = 2k$ so that n is even:

$C + \mathrm{i}S = \left(-2\mathrm{i}\left(\cos\frac{\theta}{2} + \mathrm{i}\sin\frac{\theta}{2}\right)\sin\frac{\theta}{2}\right)^{2k}$

$= (-2)^{2k}(\mathrm{i}^{2k})\left(\cos\frac{2k\theta}{2} + \mathrm{i}\sin\frac{2k\theta}{2}\right)\sin^{2k}\frac{\theta}{2}$

Compare reals: $C = (-2)^{2k}(\mathrm{i}^{2k})\cos k\theta\sin^{2k}\frac{\theta}{2}$

Compare imaginary: $S = (-2)^{2k}(\mathrm{i}^{2k})\sin k\theta\sin^{2k}\frac{\theta}{2}$

So

$\frac{S}{C} = \frac{(-2)^{2k}(\mathrm{i}^{2k})\sin k\theta\sin^{2k}\frac{\theta}{2}}{(-2)^{2k}(\mathrm{i}^{2k})\cos k\theta\sin^{2k}\frac{\theta}{2}} = \frac{\sin k\theta}{\cos k\theta}$

But $k = \frac{n}{2}$ and so $\frac{S}{C} = \frac{\sin\left(\frac{n}{2}\theta\right)}{\cos\left(\frac{n}{2}\theta\right)} = \tan\left(\frac{n}{2}\theta\right)$ as req.

41) a)

$C + \mathrm{i}S = \int_0^{\frac{\pi}{3}} \mathrm{e}^{3x} \cos 5x \, \mathrm{d}x + \mathrm{i} \int_0^{\frac{\pi}{3}} \mathrm{e}^{3x} \sin 5x \, \mathrm{d}x$

$= \int_0^{\frac{\pi}{3}} (\mathrm{e}^{3x} \cos 5x + \mathrm{i}\mathrm{e}^{3x} \sin 5x) \mathrm{d}x = \int_0^{\frac{\pi}{3}} \mathrm{e}^{3x} (\cos 5x + \mathrm{i} \sin 5x) \mathrm{d}x$

$= \int_0^{\frac{\pi}{3}} \mathrm{e}^{3x} \times \mathrm{e}^{5\mathrm{i}x} \mathrm{d}x = \int_0^{\frac{\pi}{3}} \mathrm{e}^{(3+5\mathrm{i})x} \mathrm{d}x$

$= \left[\frac{1}{3+5\mathrm{i}} \mathrm{e}^{(3+5\mathrm{i})x} \right]_0^{\frac{\pi}{3}} = \left(\frac{1}{3+5\mathrm{i}} \mathrm{e}^{\frac{\pi}{3}(3+5\mathrm{i})} \right) - \left(\frac{1}{3+5\mathrm{i}} \mathrm{e}^{0} \right) = \frac{\mathrm{e}^{\frac{\pi}{3}(3+5\mathrm{i})} - 1}{3+5\mathrm{i}}$

Since

$\mathrm{e}^{\frac{\pi}{3}(3+5\mathrm{i})} = \mathrm{e}^{\pi} \times \mathrm{e}^{\frac{5\pi}{3}\mathrm{i}} = \mathrm{e}^{\pi} \left(\cos \frac{5\pi}{3} + \mathrm{i} \sin \frac{5\pi}{3} \right) = \mathrm{e}^{\pi} \left(\frac{1}{2} - \frac{\sqrt{3}}{2} \mathrm{i} \right)$

Then

$C + \mathrm{i}S = \frac{\mathrm{e}^{\pi}\left(\frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}\right) - 1}{3+5\mathrm{i}} = \frac{\frac{1}{2}\mathrm{e}^{\pi} - \frac{\sqrt{3}}{2}\mathrm{i}\mathrm{e}^{\pi} - 1}{3+5\mathrm{i}} \times \frac{3-5\mathrm{i}}{3-5\mathrm{i}} = \frac{\frac{3}{2}\mathrm{e}^{\pi} - \frac{3\sqrt{3}}{2}\mathrm{i}\mathrm{e}^{\pi} - 3 - \frac{5}{2}\mathrm{i}\mathrm{e}^{\pi} - \frac{5\sqrt{3}}{2}\mathrm{e}^{\pi} + 5\mathrm{i}}{34}$

$= \frac{\frac{3}{2}\mathrm{e}^{\pi} - 3 - \frac{5\sqrt{3}}{2}\mathrm{e}^{\pi} + \left(5 - \frac{3\sqrt{3}}{2}\mathrm{e}^{\pi} - \frac{5}{2}\mathrm{e}^{\pi}\right)\mathrm{i}}{34} = \frac{3\mathrm{e}^{\pi} - 6 - 5\sqrt{3}\mathrm{e}^{\pi} + \left(10 - 3\sqrt{3}\mathrm{e}^{\pi} - 5\mathrm{e}^{\pi}\right)\mathrm{i}}{68}$

Compare reals:

$C = \frac{3\mathrm{e}^{\pi} - 6 - 5\sqrt{3}\mathrm{e}^{\pi}}{68} = \frac{\mathrm{e}^{\pi}}{68}\left(3 - 5\sqrt{3}\right) - \frac{3}{34}$ as req.

41) b)

Compare imaginary:

$$S = \frac{10 - 3\sqrt{3}\mathrm{e}^{\pi} - 5\mathrm{e}^{\pi}}{68} = -\frac{\mathrm{e}^{\pi}}{68}\left(5 + 3\sqrt{3}\right) + \frac{5}{34}$$

42) a)

$C + \mathrm{i}S = 1 + \cos\theta + \cos 2\theta + \cos 3\theta + \cos 4\theta + \mathrm{i}(\sin\theta + \sin 2\theta + \sin 3\theta + \sin 4\theta)$

$= 1 + (\cos\theta + \mathrm{i}\sin\theta) + (\cos 2\theta + \mathrm{i}\sin 2\theta) + (\cos 3\theta + \mathrm{i}\sin 3\theta) + (\cos 4\theta + \mathrm{i}\sin 4\theta)$

$= 1 + \mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{\mathrm{i}2\theta} + \mathrm{e}^{\mathrm{i}3\theta} + \mathrm{e}^{\mathrm{i}4\theta}$

$1 + \mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{\mathrm{i}2\theta} + \mathrm{e}^{\mathrm{i}3\theta} + \mathrm{e}^{\mathrm{i}4\theta}$ is a finite geometric series.

$a = 1$, $r = \mathrm{e}^{\mathrm{i}\theta}$ and $n = 5$

$S_n = \frac{a(1-r^n)}{1-r} = \frac{1-\mathrm{e}^{\mathrm{i}5\theta}}{1-\mathrm{e}^{\mathrm{i}\theta}}$

$\therefore C + \mathrm{i}S = \frac{1-\mathrm{e}^{\mathrm{i}5\theta}}{1-\mathrm{e}^{\mathrm{i}\theta}} = \frac{\mathrm{e}^{\frac{5}{2}\mathrm{i}\theta}}{\mathrm{e}^{\frac{1}{2}\mathrm{i}\theta}} \times \frac{\mathrm{e}^{-\frac{5}{2}\mathrm{i}\theta} - \mathrm{e}^{\frac{5}{2}\mathrm{i}\theta}}{\mathrm{e}^{-\frac{1}{2}\mathrm{i}\theta} - \mathrm{e}^{\frac{1}{2}\mathrm{i}\theta}} = \mathrm{e}^{\frac{4}{2}\mathrm{i}\theta} \times \frac{\mathrm{e}^{-\frac{5}{2}\mathrm{i}\theta} - \mathrm{e}^{\frac{5}{2}\mathrm{i}\theta}}{\mathrm{e}^{-\frac{1}{2}\mathrm{i}\theta} - \mathrm{e}^{\frac{1}{2}\mathrm{i}\theta}} = \frac{\mathrm{e}^{\frac{5}{2}\mathrm{i}\theta} - \mathrm{e}^{-\frac{5}{2}\mathrm{i}\theta}}{\mathrm{e}^{\frac{1}{2}\mathrm{i}\theta} - \mathrm{e}^{-\frac{1}{2}\mathrm{i}\theta}} \times \mathrm{e}^{2\mathrm{i}\theta}$ as req.

42) b)

Since $\mathrm{e}^{\mathrm{i}\theta} - \mathrm{e}^{-\mathrm{i}\theta} = 2\mathrm{i}\sin\theta$ then

$$\frac{2\mathrm{i}\sin\left(\frac{5}{2}\theta\right)}{2\mathrm{i}\sin\left(\frac{1}{2}\theta\right)} \times \mathrm{e}^{2\mathrm{i}\theta} = \frac{\sin\left(\frac{5}{2}\theta\right)}{\sin\left(\frac{1}{2}\theta\right)} \times (\cos(2\theta) + \mathrm{i}\sin(2\theta))$$

Real part:

$$C = \cos(2\theta) \times \frac{\sin\left(\frac{5}{2}\theta\right)}{\sin\left(\frac{1}{2}\theta\right)}$$

Since $\sin\left(\frac{5}{2}\theta\right) = \sin\left(2\theta + \frac{1}{2}\theta\right) = \sin(2\theta)\cos\left(\frac{1}{2}\theta\right) + \cos(2\theta)\sin\left(\frac{1}{2}\theta\right)$ then

$$C = \cos(2\theta) \times \frac{\sin(2\theta)\cos\left(\frac{1}{2}\theta\right) + \cos(2\theta)\sin\left(\frac{1}{2}\theta\right)}{\sin\left(\frac{1}{2}\theta\right)}$$

$$= \frac{\cos(2\theta)\sin(2\theta)\cos\left(\frac{1}{2}\theta\right) + \cos^2(2\theta)\sin\left(\frac{1}{2}\theta\right)}{\sin\left(\frac{1}{2}\theta\right)}$$

$$= \frac{\cos(2\theta)\sin(2\theta)\cos\frac{1}{2}\theta}{\sin\left(\frac{1}{2}\theta\right)} + \frac{\cos^2(2\theta)\sin\frac{1}{2}\theta}{\sin\left(\frac{1}{2}\theta\right)}$$

$$= \frac{\frac{1}{2}\sin(4\theta)\cos\left(\frac{1}{2}\theta\right)}{\sin\left(\frac{1}{2}\theta\right)} + \frac{\cos^2(2\theta)\sin\left(\frac{1}{2}\theta\right)}{\sin\left(\frac{1}{2}\theta\right)}$$

$$= \frac{1}{2}\sin(4\theta)\cot\left(\frac{1}{2}\theta\right) + \cos^2(2\theta)$$

42) c)

Since $C = \cos(2\theta) \times \frac{\sin\left(\frac{5}{2}\theta\right)}{\sin\left(\frac{1}{2}\theta\right)}$ and $S = \sin(2\theta) \times \frac{\sin\left(\frac{5}{2}\theta\right)}{\sin\left(\frac{1}{2}\theta\right)}$ then either

$\frac{\sin\left(\frac{5}{2}\theta\right)}{\sin\left(\frac{1}{2}\theta\right)} = 0$ or $\sin(2\theta) = \cos(2\theta)$

The solutions for $\frac{\sin\left(\frac{5}{2}\theta\right)}{\sin\left(\frac{1}{2}\theta\right)} = 0$ are when $\sin\left(\frac{5}{2}\theta\right) = 0$ so $\frac{2\pi}{5}$ or $\frac{4\pi}{5}$

The solutions for $\sin(2\theta) = \cos(2\theta)$ are when $\tan(2\theta) = 1$ so $\frac{\pi}{8}$ or $\frac{5\pi}{8}$

$\therefore \frac{2\pi}{5}, \frac{4\pi}{5}, \frac{\pi}{8}$ or $\frac{5\pi}{8}$

43) a)

$P = \mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{\mathrm{i}2\theta} + \mathrm{e}^{\mathrm{i}3\theta} + \mathrm{e}^{\mathrm{i}4\theta}$

$P|_{\theta=2n\pi} = \mathrm{e}^{\mathrm{i}2n\pi} + \mathrm{e}^{\mathrm{i}4n\pi} + \mathrm{e}^{\mathrm{i}6n\pi} + \mathrm{e}^{\mathrm{i}8n\pi}$

$= \left(\mathrm{e}^{\mathrm{i}2\pi}\right)^n + \left(\mathrm{e}^{\mathrm{i}2\pi}\right)^{2n} + \left(\mathrm{e}^{\mathrm{i}2\pi}\right)^{3n} + \left(\mathrm{e}^{\mathrm{i}2\pi}\right)^{4n}$

$= 1 + 1 + 1 + 1 = 4$

43) b)

$P = \mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{\mathrm{i}2\theta} + \mathrm{e}^{\mathrm{i}3\theta} + \mathrm{e}^{\mathrm{i}4\theta}$ is a finite geometric series.

$a = \mathrm{e}^{\mathrm{i}\theta}$, $r = \mathrm{e}^{\mathrm{i}\theta}$ and $n = 4$

$S_n = \frac{a(1-r^n)}{1-r} = \frac{\mathrm{e}^{\mathrm{i}\theta}(1-\mathrm{e}^{\mathrm{i}4\theta})}{1-\mathrm{e}^{\mathrm{i}\theta}}$

$\therefore P = \frac{\mathrm{e}^{\mathrm{i}\theta}(1-\mathrm{e}^{\mathrm{i}4\theta})}{1-\mathrm{e}^{\mathrm{i}\theta}} = \frac{\mathrm{e}^{\mathrm{i}\theta}}{\mathrm{e}^{\frac{1}{2}\mathrm{i}\theta}} \times \frac{1-\mathrm{e}^{\mathrm{i}4\theta}}{\mathrm{e}^{-\frac{1}{2}\mathrm{i}\theta}-\mathrm{e}^{\frac{1}{2}\mathrm{i}\theta}} = \mathrm{e}^{\frac{1}{2}\mathrm{i}\theta} \times \frac{\mathrm{e}^{\mathrm{i}4\theta}-1}{\mathrm{e}^{\frac{1}{2}\mathrm{i}\theta}-\mathrm{e}^{-\frac{1}{2}\mathrm{i}\theta}}$

Since $\mathrm{e}^{\mathrm{i}\theta} - \mathrm{e}^{-\mathrm{i}\theta} = 2\mathrm{i}\sin\theta$ then

$P = \mathrm{e}^{\frac{1}{2}\mathrm{i}\theta} \times \frac{\mathrm{e}^{\mathrm{i}4\theta}-1}{2\mathrm{i}\sin\left(\frac{1}{2}\theta\right)} = \frac{\mathrm{e}^{\frac{1}{2}\mathrm{i}\theta}(\mathrm{e}^{4\mathrm{i}\theta}-1)}{2\mathrm{i}\sin\left(\frac{1}{2}\theta\right)}$ as req.

43) c)

$P = \frac{\mathrm{e}^{\frac{1}{2}\mathrm{i}\theta}(\mathrm{e}^{4\mathrm{i}\theta}-1)}{2\mathrm{i}\sin\left(\frac{1}{2}\theta\right)} \times \frac{-\mathrm{i}}{-\mathrm{i}} = \frac{-\mathrm{i}\mathrm{e}^{\frac{1}{2}\mathrm{i}\theta}(\mathrm{e}^{4\mathrm{i}\theta}-1)}{2\sin\left(\frac{1}{2}\theta\right)} = \frac{-\mathrm{i}\mathrm{e}^{\frac{9}{2}\mathrm{i}\theta}+\mathrm{i}\mathrm{e}^{\frac{1}{2}\mathrm{i}\theta}}{2\sin\left(\frac{1}{2}\theta\right)} = \frac{-\mathrm{i}\left(\cos\left(\frac{9}{2}\theta\right)+\mathrm{i}\sin\left(\frac{9}{2}\theta\right)\right)+\mathrm{i}\left(\cos\left(\frac{1}{2}\theta\right)+\mathrm{i}\sin\left(\frac{1}{2}\theta\right)\right)}{2\sin\left(\frac{1}{2}\theta\right)}$

$= \frac{-\mathrm{i}\cos\left(\frac{9}{2}\theta\right)+\sin\left(\frac{9}{2}\theta\right)+\mathrm{i}\cos\left(\frac{1}{2}\theta\right)-\sin\left(\frac{1}{2}\theta\right)}{2\sin\left(\frac{1}{2}\theta\right)}$

$\mathrm{Re}(P) = \frac{\sin\left(\frac{9}{2}\theta\right)-\sin\left(\frac{1}{2}\theta\right)}{2\sin\left(\frac{1}{2}\theta\right)} = \frac{\sin\left(\frac{9}{2}\theta\right)}{2\sin\left(\frac{1}{2}\theta\right)} - \frac{1}{2}$ as req.

43) d)

$\frac{\sin\left(\frac{9}{2}\theta\right)}{2\sin\left(\frac{1}{2}\theta\right)} - \frac{1}{2} = 0$

$\Rightarrow \frac{\sin\left(\frac{9}{2}\theta\right)}{2\sin\left(\frac{1}{2}\theta\right)} = \frac{1}{2} \Rightarrow \sin\left(\frac{9}{2}\theta\right) = \sin\left(\frac{1}{2}\theta\right)$

Since $\sin(x) = \sin(\pi - x)$ then

$\frac{9}{2}\theta = \pi - \frac{1}{2}\theta \Rightarrow 5\theta = \pi \therefore \theta = \frac{\pi}{5}$ as req.

Since $\sin(x) = \sin(x + 2\pi)$ then

$\frac{9}{2}\theta = \frac{1}{2}\theta + 2\pi \Rightarrow 4\theta = 2\pi \therefore \theta = \frac{\pi}{2}$